

Online imbalance learning with unpredictable feature evolution and label scarcity

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ABSTRACT

Recently, online learning with imbalanced data streams has aroused wide concern, which reflects an uneven distribution of different classes in data streams. Existing approaches have conventionally been conducted on stationary feature space and they assume that we can obtain the entire labels of data streams in the case of supervised learning. However, in many real scenarios, e.g., the environment monitoring task, new features flood in, and old features are partially lost during the changing environment as the different lifespans of different sensors. Besides, each instance needs to be labeled by experts, resulting in expensive costs and scarcity of labels. To address the above problems, this paper proposes a novel Online Imbalance learning with unpredictable Feature evolution and Label scarcity (OIFL) algorithm. First, we utilize margin-based online active learning to selectively label valuable instances. After obtaining the labels, we handle imbalanced class distribution by optimizing F-measure and transforming F-measure optimization into a weighted surrogate loss minimization. When data streams arrive with augmented features, we combine the online passive-aggressive algorithm and structural risk minimization to update the classifier in the divided feature space. When data streams arrive with incomplete features, we leverage variance to identify the most informative features following the empirical risk minimization principle and continue to update the existing classifier as before. Finally, we obtain a sparse but reliable learner by the strategy of projecting truncation. We derive theoretical analyses of OIFL. Also, experiments on the synthetic datasets and real-world data streams to validate the effectiveness of our method.

1. Introduction

Due to the openness and complexity of the real environment, in many practical applications, people have to face the situation that data have imbalanced class distributions, such as cancer diagnosis [1], financial credit monitoring [2] and abnormalities [3]. In such cases, the performance will be limited by the size of the minority training sample. The accuracy may be misleading, as the model may achieve high accuracy on the majority classes but perform poorly on the minority classes. Also, it would lead to misclassifications, as the model might incorrectly classify the minority classes as majority classes due to their smaller sample sizes, resulting in insufficient learning of their features. We are always concerned with the performance of learning algorithms in the class with rare events. As data streams arrive, each instance should be labeled by experts, which leads to expensive costs and the scarcity of labels. But, it is not always informative or necessary to query the labels of each instance. There are a few online learning algorithms that have been developed to learn from imbalanced data [4–6]. However, they all assume that they query the label of each instance and that data streams have a fixed, stable, and predefined feature space, which is unrealistic and costly in many real scenarios.

For example, in the environment monitoring task of deserts, several sensors are installed over the specific environment to collect data about humidity, temperature, PH, and so on, where sensors collect much more data about dryness than wetness and we selectively query valuable instances to be labeled. Due to the different conditions of the sensors, such as the difference in position, temperature, signal range, etc., the expiring time of the sensors will be different and unpredictable. To maintain the observation of the environment before the short-lived sensors expire, new sensors are installed before the roughly estimated out-of-service time of all sensors on the environment so that the data contains historical features and new features for a short period. Note that the position or the number of the new sensors could be different from the old ones. Therefore, as shown in Fig. 1, features corresponding to the expired sensors suffer from feature loss, and features received by new sensors are inherited.

Online imbalance learning with unpredictable feature evolution and label scarcity faces more difficulties than existing online learning problems because we try to learn as much information as possible

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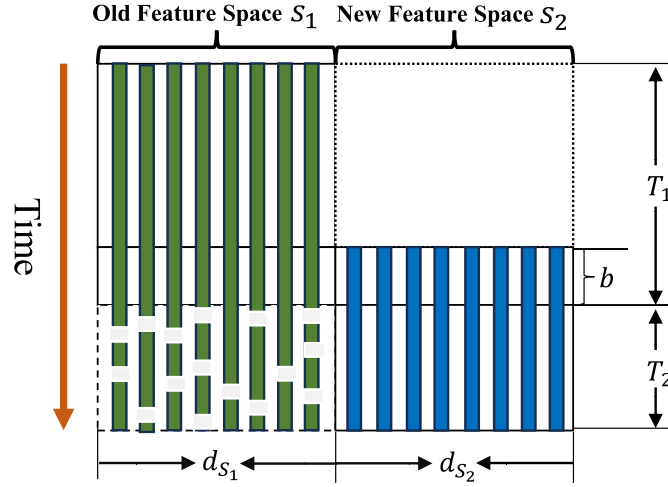


Fig. 1. Illustration of how imbalanced data streams arrive. Each column of different colors denotes the features collected from the sensors. At the beginning $1, \dots, T_1 - b$, only feature space S_1 is observed. For $t \in [T_1 - b + 1, T_1]$, the feature set S_2 corresponding to the new sensors appears. After period T_1 , some old sensors start to wear out and S_1 loses part features.

from dynamic feature spaces and imbalanced data streams with minimal supervision. There are three essential challenges in this learning paradigm. The first one is how to relax the restriction that all instances must be labeled and how to select informative instances. The second one is how to alleviate the limitations of imbalanced data distribution and how to effectively extract useful information from instances of different classes to build better classifiers, where we may pay more attention to minority classes [7,8]. The last one is how to design a model update strategy with highly dynamic feature space, so that the model can adapt to the unpredictable feature evolution process, mining more useful information. For the first challenge, a series of studies about online active learning has been explored to learn from data streams with scarce labels [9–12]. For the second challenge, there are a series of studies, such as online classification based on cost-sensitive [13–16] and online bagging and boosting for imbalanced data streams [17], which are explored in stationary and invariable feature space. For the last challenge, another series of studies are proposed to relax the constraint that the feature space of all incoming instances must be fixed [18–22].

Nevertheless, the model updating criteria of the above imbalance learning algorithms always accompanied by predefined assumptions, no existing online imbalance learning can fully characterize the unpredictable feature evolution and label scarcity. Motivated by these problems, we propose a novel paradigm: Online Imbalance learning with unpredictable Feature evolution and Label scarcity (OIFL). Our goal is to fully and efficiently exploit the information of data streams with unpredictable feature evolution from imbalanced data and build a powerful predictive model with minimal supervision. After labeling informative instances by active learning strategy, we first optimize the F-measure and transform F-measure optimization into a weighted surrogate loss minimization to alleviate the influence of imbalanced class distribution. Then, we combine the online passive-aggressive algorithm and structural risk minimization to update the classifier in divided feature space as data streams arrive with augmented features. When data streams arrive with incomplete features, we utilize variance to identify the most informative features following the empirical risk minimization principle and continue to update the existing classifier as before. Finally, we use the projected truncation technique to build a sparse but efficient model. Theory analysis and experimental exploration validate the effectiveness of our method.

Generally speaking, we can conclude the main contributions as follows:

- (1) We propose a new algorithm OIFL to deal with unpredictable feature evolution learning with imbalanced data streams based on minimal supervision, in which the volume and feature dimension of incoming data streams change unpredictably as well as label scarcity in case of imbalanced data distribution.
- (2) We handle imbalanced data distribution by optimizing F-measure. Then, we combine the online passive-aggressive algorithm and the adaptive weighted strategy to deal with dynamic feature spaces. Moreover, we analyze the error bounds theoretically.
- (3) Experiments on the synthetic datasets and real-world data streams validate the effectiveness of our method.

The rest of this article is organized as follows. The Section 2 introduces the related work. The formulation of the proposed algorithm is presented in Section 3. The Section 4 provides the corresponding theoretical analysis. The experimental results are shown in Section 5, and the conclusions are shown in Section 6.

2. Related work

The classification model is mainly learned for unpredictable feature evolution with imbalanced data streams, which is mainly related to online learning of dynamic feature space and imbalance learning. We review the related work in this paper mainly from the two aspects mentioned above.

2.1. Online learning from dynamic feature space

The dynamic feature space means feature space can change or evolve when learning with streaming data. A typical example is trapezoidal data streams [18], where both volume and feature dimensions of data may simultaneously expand over time. The most relevant to our work is FLLS [22], which is proposed to handle trapezoidal data streams with label scarcity. It designs a margin-based online active learning strategy to label valuable instances and then build superior predictive models with minimal supervision. Another most relevant work is FESL [21], where they find that historical features would vanish along with new features arriving. Specifically, when previous features vanish, they simulate them by a mapping function learned in the overlapping period that contains old features and new features. And then, ensemble learning is used to integrate two models that learn from the above two parts, respectively. PUFE [20] has made an extension about the above scenario, it proposes a matrix completion technique

to deal with the case that there will be no integrally overlapping period since different features could vanish unpredictably. Also, based on FESL, EDM [23] makes a further exploration that it designs a discrepancy measure across two different feature spaces and provides effective algorithms with sound theoretical guarantees. Learning with more complex feature space OLVF [19] and GLSC [24] is also relevant to our work, where they assume that the features of incoming instances would lose and appear arbitrarily.

In comparison, our OIFL algorithm has strategies to deal with imbalanced class distributions that are not considered by the above algorithms. Besides, our OIFL algorithm does not make any assumptions in the feature evolution scenario, which makes it not subject to the limitations of the related algorithms.

2.2. Imbalance learning

Recently, the imbalance learning problems have drawn significant interest from the machine learning and data mining field. The first challenge in imbalance learning is on account of the dynamic or non-stationary nature of data streams in which the underlying distribution of data is changing over time violently called concept drift [25]. Another challenge is that imbalanced data can significantly influence the performance of most standard learning algorithms [26], the main reason is that a trivial learner that simply classifies any instance as negative can obtain higher accuracy in the case that the negative class is the majority class. Handling the imbalance problem with concept drift becomes more profound and significant which leads to many existing surveys focusing on them simultaneously. Following are a few state-of-the-art solutions to alleviate the above problem and we put them into three categories. The first part is the data-level methods [27–30], which are the most widely used methods that modify the underlying data to make its distribution more balanced to improve the classifier's accuracy by resampling over the dataset. The most popular existing algorithm-level approach is cost-sensitive learning, which assigns a higher misclassification cost to the minority while a smaller misclassification cost is assigned to the majority sample. For instance, Adel Ghazikhani et al. [31] designed a two-layer approach for handling class imbalance and non-stationarity. In the first layer, cost-sensitive learning is embedded into the training phase of the NNs, and in the second layer, a new method for weighting classifiers of the ensemble is proposed. TSCS [32] proposes a novel two-stage cost-sensitive framework for data stream classification by utilizing cost information in both the feature selection stage and classification stage. The third ensemble learning class methods focus on combining a data-level or algorithm-level approach with ensemble learning to obtain powerful ensemble classifiers, such as OOB and UOB [33], AREBA [34], and ROSE [35].

However, the above algorithms always assume that we can fully knowledge label information in the stationary and completed data stream spaces, which makes them unattractive for mining large amounts of evolving data streams as it is costly to query all labels. Conversely, our OIFL algorithm aims to handle learning from imbalanced data streams with unpredictable feature evolution and label scarcity that cannot be handled by the above-related algorithms.

3. Formulation

In this section, we introduce the general problem settings. Besides, we provide the details of our OIFL and the optimization problem-solving.

3.1. Problem setting

Without loss of generality, we consider a task of online binary classification problems with imbalanced data and feature evolution here. As shown in Fig. 1, the feature evolution of the data shows an obvious phased trend in the scenario. Thus, we divide the feature evolution processing into three stages and data have relatively stable features in different stages. At various stages, we adopt active learning strategies to decide whether to acquire the incoming instance. In general, we can summarize the rules of feature evolution as:

- ① $1 \sim T_1 - b_1$
The feature space of the data during this time is the same as the first arrived instance x_1 , which only contains the feature set S_1 .
- ② $T_1 - b_1 + 1 \sim T_1$
At the beginning of the phase, $t = T_1 - b_1 + 1$, the mutation occurs in the feature space that the volume and feature dimension of data streams simultaneously increase. Data carries features from both S_1 and S_2 at the moment which is called the transition node.
- ③ $T_1 + 1 \sim T_2$
At this stage, feature reduction randomly of data streams compared to the previous stage, where S_2 gets inherited and S_1 loses part features.

Let data streams $D = \{(x_t, y_t) | t \in \{1, 2, \dots, T\}\}$ denote a sequence of arrival training instances, where $x_t \in \mathbb{R}^{d_t}$ represents a vector have d_t dimensions at time t and $y_t \in \{-1, +1\}$ represents the true label of x_t . For specifically, at round t , an instance x_t is arriving and the learner predicts its estimated class label as $\hat{y}_t = \text{sign}(w_t \cdot x_t)$, in which $w_t \in \mathbb{R}^{d_{t-1}}$ represents a linear predictive classifier learned from previous $t - 1$ instances and $w_1 \in \mathbb{R}^{d_1}$ is an initialized zero vector. Then, the model acquires the real label of the instance $y_t \in \{-1, +1\}$. If $\hat{y}_t \neq y_t$, it means the learner made a wrong prediction and suffered instantaneous losses. We list the notions used first in our paper as well as their meanings in Table 1.

For the measurement indicators of the online imbalance binary classification task, we first assume the positive examples as a rare class. Traditional online learning tends to optimize and maximize classification accuracy. This may be inappropriate for imbalanced data because a trivial learner could still achieve a high accuracy by simply classifying all examples as negative that represents the majority class. In the presence of imbalanced classes, our goal is to minimize the number of misclassified instances on all datasets. Thus, a more appropriate approach is to maximize the F-measure based on minimizing structural and empirical risk [36,37].

$$F_\beta = \frac{(1 + \beta^2)TP}{\beta^2(TP + FN) + TP + FP} \quad (1)$$

where TP, FN denotes the number of instances predicted to be positive and negative, respectively, while the actual class is positive. TN, FP denotes the number of instances predicted as negative and positive, respectively, when the actual class is negative. Then, we can adjust our attention to maximize the F-measure.

3.2. The proposed algorithm

In this subsection, we first provide an online learning framework for imbalance learning with label scarcity from 1 to T_1 . Then, for $T_1 + 1 \sim T_2$, another adaptive weighted strategy is proposed to deal with incomplete data streams.

3.2.1. Stage T_1 : Learning from imbalanced data streams with label scarcity

The feature evolution of data presents obvious rules under this scenario. For $1 \sim T_1$, feature space increases at $t = T_1 - b + 1$ and remains steady for the rest time. We divide the projection of x_t onto the shared feature space of S_1 as x_t^s and the new feature space of S_2 as x_t^n . Thus, the role of w_t^s and w_t^n is to update existing features of S_1 and

Table 1

Notations.

Notations	Descriptions	Notations	Descriptions
y_i	$y_i \in \{-1, 1\}$, true label of x_i	$\hat{y}_i$	$\hat{y}_i \in \{-1, 1\}$, predicted label of x_i
$\mathbf{w}_t$	$\mathbf{w}_t \in \mathbb{R}^{d_t-1}, \mathbf{w}_1 \in \mathbb{R}^{d_1}$, classifier built at iteration $t-1$	C	$C > 0$, penalty cost parameter
ℓ_i	Hinge loss of instance (x_i, y_i)	$\ell_i^*(c)$	Hinge loss on instance (x_i, y_i) for any vector $c \in \mathbb{R}^{d_t}$
$\mathbf{x}_i^s$	A vector of shared features in S_1	$\mathbf{x}_i^d$	A vector of disappeared features in S_1
$\mathbf{x}_i^n$	A vector of new incoming features from S_2	τ_i	Learning rate variable
γ_i	The cost of the instance belongs to the class of iteration t	σ	The scale factor of classification cost
T	Representing total numbers of obtained instances	A_t	$A_t \in \{0, 1\}$, Bernoulli random variable
ξ	Slack variable	B	Residual feature ratio after truncation
$\Omega(y_i)$	An activation function	μ_i^i	The informativeness of the i th feature in instance x_i
m_i^s	Confidence of features on the shared feature space of S_1	m_i^n	Confidence of features on the new feature space of S_2
$\mathbf{w}_i^d$	The projection of $\mathbf{w}_i$ onto disappear feature space, i.e., $\Pi_{x_i^d} \mathbf{w}_i$	$\mathbf{w}_i^s$	The projection of $\mathbf{w}_i$ onto shared feature space, i.e., $\Pi_{x_i^s} \mathbf{w}_i$
$\mathbf{w}_i^n$	The projection of $\mathbf{w}_i$ onto new feature space, i.e., $\Pi_{x_i^n} \mathbf{w}_i$	$\bar{\mathbf{w}}_{t+1}$	Transition variable of $\mathbf{w}_{t+1}$ after updating process
μ_i^i	The informativeness of i th feature in instance x_i	δ	$\delta > 0$, a smoothing parameter
T_P	The number of positive (minority) instance	T_N	The number of negative (majority) instances
$\hat{y}_i$	Predicted class label	p_i	The prediction confidence
TP, FN	The number of instances predicted to be positive and negative, respectively, while the actual class is positive	TN, FP	The number of instances predicted as negative and positive, respectively, when the actual class is negative

S_2 , respectively. To mitigate the impact of label scarcity in iteration, our algorithm decides whether to query the correctness of the label for each instance x_i arrived at round t based on a Bernoulli random variable $A_t \in \{0, 1\}$ with probability:

$$q_t = \frac{\delta}{\delta + |p_t|} \quad (2)$$

$$p_t = \Pi_{x_t} \mathbf{w}_t \cdot \mathbf{x}_t$$

where p_t can be seen as prediction confidence and $\delta > 0$ is a smooth parameter that trades off the ratio of queries. Such an approach is inspired by margin-based active learning [38,39] and has wide applications [22,40,41]. We use p_t to represent the importance of instance in the existing classifier. It can influence our decisions with a certain probability to avoid directly querying the label and making a wrong decision when there are not enough instances to try a reliable learner in the early time. There are two cases for A_t , if $A_t = 0$, the algorithm OIFL will not query the class label of instance, and the model remains untouched. If $A_t = 1$, the class label is queried and the OIFL will make an instantaneous loss $\ell_t(x_i, \mathbf{w}_t; y_i)$ between $\hat{y}_i$ and y_i , where $y_i \in \{-1, 1\}$. Then, OIFL will adopt the proposed updating step to update the new linear classifier $\mathbf{w}_{t+1}$. Before presenting our algorithms, we prove an important proposition below for imbalance learning.

Proposition 1. *Considering an imbalance classification problem, the goal of maximizing the F-measure Eq. (1) is equivalent to minimizing the following objective:*

$$\gamma_{10} \sum_{i=1}^{T_P} \mathbb{I}(y_i \neq \text{sign}(\mathbf{w}_t \cdot \mathbf{x}_i) \mid y_i = +1)) \quad (3)$$

$$+ \gamma_{01} \sum_{i=1}^{T_N} \mathbb{I}(y_i \neq \text{sign}(\mathbf{w}_t \cdot \mathbf{x}_i) \mid y_i = -1)).$$

where γ_{10}, γ_{01} indicate the importance(weight), $\mathbb{I}(\cdot)$ is an indicator function. T_P, T_N denote the number of positive(minority) and negative(majority) instances at training data set.

Proof. By analyzing the function of the F_β in Eq. (1) and replacing TP by $T_P - FN$ we can derive the following conclusions:

$$\max F_\beta \Rightarrow \min[1 + \frac{\beta^2 FN + FP}{(1 + \beta^2)(T_P - FN)}] \quad (4)$$

$$\Rightarrow \min[\beta^2 FN + FP - (1 + \beta^2)(T_P - FN)]$$

$$\Rightarrow \min[(1 + 2\beta^2)FN + FP - (1 + \beta^2)T_P].$$

where $(1 + \beta^2)T_P$ is a constant. From Eq. (4), we can find that maximizing F_β -measure is equivalent to

$$\min[(1 + 2\beta^2)FN + FP]. \quad (5)$$

in the task of imbalance data streams, losses borne by minority misclassification always suffer more than those of the majority, i.e. FN shows more importance than FP . Moreover, learners not only focus more on the minority but also want to reduce the number of misclassified instances of these two classes and obtain optimizer metrics. Thus, we can transform Eq. (5) into a weighted form and define it as:

$$\min[\gamma_{10} FN + \gamma_{01} FP]$$

$$= \min[\gamma_{10} \sum_{i=1}^{T_P} \mathbb{I}(y_i \neq \text{sign}(\mathbf{w}_t \cdot \mathbf{x}_i) \mid y_i = +1) \quad \square \quad (6)$$

$$+ \gamma_{01} \sum_{i=1}^{T_N} \mathbb{I}(y_i \neq \text{sign}(\mathbf{w}_t \cdot \mathbf{x}_i) \mid y_i = -1)].$$

Proposition 1 gives an explicit objective function for optimization, but the indicator function is non-convex and inconvenient for direct optimization. To effectively solve the above problem, we choose hinge loss as the loss function,

$$\ell_i = \ell_i(\mathbf{x}_i, \mathbf{w}_t; y_i) = \max\{0, (1 - y_i(\mathbf{w}_t \cdot \mathbf{x}_i))\}, \quad (7)$$

which has been widely used and proven effective in many real-world applications [5,22,42]. Thus we convert the optimization problem that gets the number of misclassifications to minimize the surrogate function—the hinge loss. Therefore, Eq. (3) can be expressed as:

$$\min[\gamma_{10} \sum_{i=1}^{T_P} \ell_i(\mathbf{x}_i, \mathbf{w}_t; y_i = +1) + \gamma_{01} \sum_{i=1}^{T_N} \ell_i(\mathbf{x}_i, \mathbf{w}_t; y_i = -1)], \quad (8)$$

$$= \min[\gamma_t \sum_{i=1}^T \ell_i(\mathbf{x}_i, \mathbf{w}_t; y_i)].$$

Since only one instance can be obtained in iteration t , we replace γ_{10} and γ_{01} used by γ_t , which indicates the cost of the instance belongs to the class of iteration t . To make the learning steps controllable, we introduce a scale factor σ to adjust the convergence speed of the learning algorithm [5]. We get the expression of cost function γ_t :

$$\gamma_t = \frac{\sigma}{(\frac{T_P}{T_N})\Omega(y_t) + (\frac{T_N}{T_P})(1 - \Omega(y_t))}, \quad (9)$$

where $\Omega(y_t)$ is an activation function based on y_t , which is defined as:

$$\Omega(y_t) = \begin{cases} 1, & \text{if } y_t = 1 \\ 0, & \text{if } y_t = -1 \end{cases}, \quad (10)$$

therefore, we can get a modified hinge loss function for each obtained x_i :

$$\gamma_t \ell_i(\mathbf{x}_i, \mathbf{w}_t; y_i) = \max\{0, \gamma_t(1 - y_i(\mathbf{w}_t \cdot \mathbf{x}_i))\}. \quad (11)$$

We use the online passive-aggressive learning approach proposed in [18,43] to online update $\mathbf{w}_t$ with each instance $(\mathbf{x}_t, y_t)$,

$$\bar{\mathbf{w}}_{t+1} = \arg \min_{\substack{\mathbf{w} \in [\mathbf{w}^s, \mathbf{w}^n] \\ \gamma_t \ell_t(\mathbf{x}_t, \mathbf{w}; y_t) > 0}} \frac{1}{2} \|\Pi_{x_t} \mathbf{w} - \Pi_{x_t} \mathbf{w}_t\|^2 + C\xi, \quad (12)$$

$$s.t. \gamma_t \ell_t(\mathbf{x}_t, \mathbf{w}_t; y_t) = \max\{0, \gamma_t(1 - y_t(\Pi_{x_t} \mathbf{w} \cdot \mathbf{x}_t))\}.$$

where $t \in [1, T_1]$, the slack variable ξ is used to alleviate overfitting and poor generalization caused by noise. $\mathbf{w}_1 \in \mathbb{R}^{d_{S_1}}$ is the initial zero vector. $\mathbf{w}^s \in \mathbb{R}^{d_{S_1}}$ and $\mathbf{w}^n \in \mathbb{R}^{d_{S_2}}$ are the projection of $\mathbf{w}$ onto S_1 and S_2 respectively. When $t = T_1 - b + 1$, $\mathbf{w}^s = \mathbf{w}_t$, $\mathbf{w}^n = (0, \dots, 0)$ and $\mathbf{w}_{t+1} = [\mathbf{w}_t, 0, \dots, 0]$. Eq. (12) tends to retain as much knowledge as possible learned from arrived instances based on the empirical risk minimization principle. In addition, we can deduce that the solution to the above problem is the projection of $\mathbf{w}$ onto the set of all disjoint weight vectors, in which we can obtain the prediction loss. Specially, if $\hat{y}_t = y_t$ for existing $\mathbf{x}_t$, it means that the precision of current $\mathbf{w}$ for incoming $\mathbf{x}_t$ is right, i.e. $\gamma_t \ell_t(\mathbf{x}_t, \mathbf{w}; y_t) = \max\{0, \gamma_t(1 - y_t(\Pi_{x_t} \mathbf{w} \cdot \mathbf{x}_t))\} = 0$. The updated algorithm is *passive*. Meanwhile, if $\hat{y}_t \neq y_t$, classifier $\mathbf{w}_t$ makes mistakes in label prediction of arrived $\mathbf{x}_t$, i.e., $\gamma_t \ell_t(\mathbf{x}_t, \mathbf{w}_t; y_t) = \max\{0, \gamma_t(1 - y_t(\Pi_{x_t} \mathbf{w}_t \cdot \mathbf{x}_t))\} > 0$. The algorithm *aggressively* updates classifier $\mathbf{w}_t$ utilizing modified hinge loss.

3.2.2. Stage T_2 : Learning from incomplete feature space

With incomplete data streams arrive, for $T_1 + 1 \sim T_2$, the absence of features appears in S_1 , and S_2 remains stable. To adapt to the tendency of feature evolution and make an accurate prediction, our algorithm utilizes an adaptive weighted strategy based on uncertainty to learn from feature space S_1 and S_2 . Especially, at each round t , $t \in (T_1, T_2]$, when $\mathbf{x}_t$ arrives with old features dissipating, we denote the projection of $\mathbf{x}_t$ onto the dissipated feature space of S_1 as $\mathbf{x}_t^d$. Thus, the updated classifier $\mathbf{w}_t$ will be divided into three parts, which can be written by $[\mathbf{w}_t^d, \mathbf{w}_t^s, \mathbf{w}_t^n]$, as the projection of $\mathbf{w}_t$ on the $[\mathbf{x}_t^d, \mathbf{x}_t^s, \mathbf{x}_t^n]$, where $\mathbf{w}_t^d, \mathbf{w}_t^s \in \mathbb{R}^{d_{S_1}}$ and $\mathbf{w}_t^n \in \mathbb{R}^{d_{S_2}}$.

In our work, we replace uncertainty with variance to estimate the informativeness of a feature [44,45], since the fact that the variance of a feature is large enough during the process of iteration, means that it contains more uncertainty that the feature would provide more information to update the model. Besides, the disappeared features cannot provide any useful information for updating the model. Concretely, the cumulative value of the informativeness from features is regarded as the confidence of the feature space after projecting $\mathbf{x}_t$ onto different feature spaces. We denote the informativeness of the i th feature in instance $\mathbf{x}_t$ by μ_t^i and represent the confidence of shared feature space by m_t^s , which is defined as:

$$m_t^s = \sum_{i=1}^{d_t^s} \frac{\mu_t^i}{\sum_{j=1}^{d_t^s} \mu_t^j}, \quad (13)$$

where d_t^s is the dimension of the shared feature space. Similarly, the notion m_t^n represents the confidence of the new feature space. We train the classifier $\mathbf{w}_t$ by modifying the hinge loss according to the weights of different spaces [5]. Thus, following the *passive-aggressive* learning method, the update of $\mathbf{w}_t$ can be transformed into the update of $\mathbf{w}_t^s$ and $\mathbf{w}_t^n$ by optimizing the online objective function:

$$\bar{\mathbf{w}}_{t+1} = \arg \min_{\substack{\mathbf{w} = [\mathbf{w}_t^d, \mathbf{w}_t^s, \mathbf{w}_t^n] \\ \gamma_t \ell_t \leq \xi, \xi \geq 0}} \frac{1}{2} \|\mathbf{w}^d - \mathbf{w}_t^d\|^2 + \frac{1}{2} \|\mathbf{w}^s - \mathbf{w}_t^s\|^2 + \frac{1}{2} \|\mathbf{w}^n - \mathbf{w}_t^n\|^2 + C\xi, \quad (14)$$

$$s.t. \gamma_t \ell_t = \max\{0, \gamma_t(1 - y_t(m_t^s \cdot \mathbf{w}^s \cdot \mathbf{x}_t^s) - y_t(m_t^n \cdot \mathbf{w}^n \cdot \mathbf{x}_t^n))\}.$$

where $t \in (T_1, T_2]$. When the learner obtains an instance, it will reassign the weights of different feature Spaces based on the feature confidence, the more important feature space will be given greater weight. Also, we focus on the feature set S_2 , which is on behalf of the tendency of feature evolution.

Due to the expansion of the feature space, all features are stored in the classifier, which may lead to high calculation and storage costs. To improve the running efficiency, we remove redundant features by the projection of $L1$ ball and truncation based on $B \in (0, 1]$ and retain relatively important ones [22]. Usually, we utilize $\text{floor}(B \cdot d_t)$ instead of $B \cdot d_t$, which may not be integer. The projection is as follows:

$$\mathbf{w}_{t+1} = \min\{1, \frac{\epsilon}{\|\bar{\mathbf{w}}_{t+1}\|_1}\} \bar{\mathbf{w}}_{t+1}, \quad (15)$$

where $\epsilon > 0$ is the regularization parameter. By using this strategy, we get the final classifier $\mathbf{w}_{t+1}$ that removes the redundancy.

3.3. Optimization

We utilize the Lagrangian function and K.K.T. conditions [46] to solve the optimization function with inequality constraints. Thus, when $t \in [1, T_1]$, we obtain the Lagrangian function of Eq. (12):

$$\mathcal{L}(\mathbf{w}, \xi, \tau, \lambda) = \frac{1}{2} \|\mathbf{w}^s - \mathbf{w}_t^s\|^2 + \frac{1}{2} \|\mathbf{w}^n - \mathbf{w}_t^n\|^2 + C\xi + \tau(\gamma_t \ell_t - \xi) - \lambda\xi, \quad (16)$$

where $\tau \geq 0$, $\lambda \geq 0$ are Lagrange multiplier. Let us take the derivative of $\mathcal{L}(\mathbf{w}, \xi, \tau, \lambda)$ concerning $\mathbf{w}_t^s, \mathbf{w}_t^n, \tau, \lambda$, and make them equal to zero, getting the following results:

$$\begin{aligned} \nabla_{\mathbf{w}_t^s}(\mathcal{L}) &= \mathbf{w}^s - \mathbf{w}_t^s - \tau \gamma_t y_t \mathbf{x}_t^s = 0 \Rightarrow \mathbf{w}^s = \mathbf{w}_t^s + \tau \gamma_t y_t \mathbf{x}_t^s, \\ \nabla_{\mathbf{w}_t^n}(\mathcal{L}) &= \mathbf{w}^n - \mathbf{w}_t^n - \tau \gamma_t y_t \mathbf{x}_t^n = 0 \Rightarrow \mathbf{w}^n = \mathbf{w}_t^n + \tau \gamma_t y_t \mathbf{x}_t^n, \\ \nabla_{\xi}(\mathcal{L}) &= C - \tau - \lambda = 0 \Rightarrow \lambda = C - \tau. \end{aligned} \quad (17)$$

Plugging the above back into Eq. (16) we get,

$$\begin{aligned} \mathcal{L}(\tau) &= -\frac{1}{2} \tau^2 \gamma_t^2 \|\mathbf{x}_t^s\|^2 - \frac{1}{2} \tau^2 \gamma_t^2 \|\mathbf{x}_t^n\|^2 + \tau \gamma_t (1 - y_t(\mathbf{w}^s \cdot \mathbf{x}_t^s) - y_t(\mathbf{w}^n \cdot \mathbf{x}_t^n)), \\ \nabla_{\tau}(\mathcal{L}) &= \tau \gamma_t (\|\mathbf{x}_t^s\|^2 + \|\mathbf{x}_t^n\|^2) - (1 - y_t(\mathbf{w}^s \cdot \mathbf{x}_t^s) - y_t(\mathbf{w}^n \cdot \mathbf{x}_t^n)) = 0, \\ \Rightarrow \tau_t &= \min\{C, \frac{1 - y_t(\mathbf{w}^s \cdot \mathbf{x}_t^s) - y_t(\mathbf{w}^n \cdot \mathbf{x}_t^n)}{\gamma_t (\|\mathbf{x}_t^s\|^2 + \|\mathbf{x}_t^n\|^2)}\}, \\ &= \min\{C, \frac{\ell_t}{\gamma_t \|\mathbf{x}_t\|^2}\}. \end{aligned} \quad (18)$$

the update rule is derived as follows:

$$\bar{\mathbf{w}}_{t+1} = [\mathbf{w}_t^s + \tau_t \gamma_t y_t \mathbf{x}_t^s, \mathbf{w}_t^n + \tau_t \gamma_t y_t \mathbf{x}_t^n]. \quad (19)$$

Meanwhile, for $t \in (T_1, T_2]$, the expression of online objective function as Eq. (16), similar to the above process, we have

$$\begin{aligned} \tau_t &= \min\{C, \frac{\ell_t}{\gamma_t [(m_t^s)^2 \|\mathbf{x}_t^s\|^2 + (m_t^n)^2 \|\mathbf{x}_t^n\|^2]}\}, \\ \bar{\mathbf{w}}_{t+1} &= [\mathbf{w}_t^d, \mathbf{w}_t^s + \tau_t m_t^s \gamma_t y_t \mathbf{x}_t^s, \mathbf{w}_t^n + \tau_t m_t^n \gamma_t y_t \mathbf{x}_t^n] \\ &= [\mathbf{w}_t^d, \mathbf{w}_t^s + \min\{C \gamma_t m_t^s y_t \mathbf{x}_t^s, \frac{\ell_t m_t^s y_t \mathbf{x}_t^s}{((m_t^s)^2 \|\mathbf{x}_t^s\|^2 + (m_t^n)^2 \|\mathbf{x}_t^n\|^2)}\}, \\ &\quad \mathbf{w}_t^n + \min\{C \gamma_t m_t^n y_t \mathbf{x}_t^n, \frac{\ell_t m_t^n y_t \mathbf{x}_t^n}{((m_t^s)^2 \|\mathbf{x}_t^s\|^2 + (m_t^n)^2 \|\mathbf{x}_t^n\|^2)}\}]. \end{aligned} \quad (20)$$

Finally, the general procedure of the method is presented in Algorithm 1.

4. Theoretical analysis

We give the theoretical analysis of the proposed algorithm OIFL in this section. Below put forward two theorems to demonstrate the performance metrics of our algorithm. Before proposing the upper error bounds, we show a lemma first, which is convenient for the proofs. For facility, we present the notions: $\mathcal{R} = \{t \mid t \in [T], \hat{y}_t \neq y_t\}$, and $\mathcal{H} = \{t \mid t \in [T], \hat{y}_t = y_t, \ell_t(\mathbf{x}_t, \mathbf{w}_t; y_t) > 0\}$, where $[T] \in \{1, 2, \dots, T\}$.

¹ T_1 represents the transitional moment between maintaining feature integrity and randomly reducing features in S_1 .

Algorithm 1 OIFL Algorithm

Input: smooth parameter $\delta > 0$, penalty parameter $C > 0$, proportion of selected features $B \in (0, 1]$, The scale factor of classification cost $\xi > 0$, regulation parameter $\lambda > 0$, classification cost $\gamma_t > 0$

- 1: Initialize: $\mathbf{w}_1 = (0, \dots, 0) \in \mathbb{R}^{d_1}$
- 2: **for** $t = 1, 2, \dots, T$ **do**
- 3: obtain $\mathbf{x}_t \in \mathbb{R}^{d_t}$
- 4: project $\mathbf{w}_t, \mathbf{x}_t$ onto $\mathbb{R}^s, \mathbb{R}^n$, respectively: $\mathbf{w}_t^s, \mathbf{w}_t^n, \mathbf{x}_t^s, \mathbf{x}_t^n$
- 5: set $p_t = \mathbf{w}_t \cdot \Pi_{\mathbf{w}_t} \mathbf{x}_t$, predict $\hat{y}_t = \text{sign}(p_t)$
- 6: update the confidence of features by variance
- 7: draw a Bernoulli random variable $\Lambda_t \in \{0, 1\}$ of probability $\delta/(\delta + |p_t|)$
- 8: **if** $\Lambda_t = 1$ **then**
- 9: query the label of $\mathbf{x}_t$: $y_t \in \{-1, 1\}$
- 10: **if** $t \leq T_1$ **then**
- 11: modified suffer loss: $\gamma_t \ell_t = \max\{0, \gamma_t(1 - y_t(\mathbf{w}_t^s \cdot \mathbf{x}_t^s) - y_t(\mathbf{w}_t^n \cdot \mathbf{x}_t^n))\}$
- 12: compute τ_t according to Eq. (18)
- 13: update classifier with $\gamma_t, y_t, \mathbf{x}_t^s, \mathbf{x}_t^n, C, \ell_t$ using Eq. (19)
- 14: **else**
- 15: calculate the confidence m_t^s, m_t^n of features on different feature spaces using Eq. (13)
- 16: modified suffer loss $\gamma_t \ell_t = \max\{0, \gamma_t(1 - y_t(m_t^s \cdot \mathbf{w}_t^s \cdot \mathbf{x}_t^s) - y_t(m_t^n \cdot \mathbf{w}_t^n \cdot \mathbf{x}_t^n))\}$
- 17: compute τ_t similar to Eq. (20)
- 18: update classifier with $m_t^s, m_t^n, \gamma_t, y_t, \mathbf{x}_t^s, \mathbf{x}_t^n, C, \ell_t$ using Eq. (20)
- 19: **end if**
- 20: project $\bar{\mathbf{w}}_{t+1}$ to a L_1 ball using Eq. (15) with ε
- 21: truncate classifier based on B and obtain $\mathbf{w}_{t+1}$
- 22: **else**
- 23: $\mathbf{w}_{t+1} = \mathbf{w}_t$
- 24: **end if**
- 25: **end for**

Lemma 4.1. Let $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_T, y_T)$ be a sequence of input instances, where $\mathbf{x}_t \in \mathbb{R}^{d_t}, d_{t-1} \leq d_t, y_t \in \{-1, +1\}, t = 1, 2, \dots, T, \gamma_t = \frac{\sigma}{(\frac{n_+}{n_-})\Omega(y_t) + (\frac{n_-}{n_+})(1 - \Omega(y_t))}$, Let τ_t be the step size for OIFL as given in Eqs. (18) and (20). The following bounds hold for any $\mathbf{c} \in \mathbb{R}^{d_T}$.

$$\begin{aligned} \sum_{t=1}^T 2\Lambda_t \tau_t \gamma_t [H_t(\kappa - |p_t|) + R_t(\kappa + |p_t|)] &\leq \kappa^2 \|\mathbf{c}\|^2 \\ &+ \sum_{t=1}^T \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 + \sum_{t=1}^T 2\kappa \tau_t \gamma_t \ell_t^*(\mathbf{c}), \end{aligned} \quad (21)$$

where $H_t = \mathbb{I}_{t \in \mathcal{H}}, R_t = \mathbb{I}_{t \in \mathcal{R}}, \mathbb{I}$ is an indicator function, and $\kappa > 0$. $\ell_t^*(\mathbf{c}) = \max\{0, 1 - y_t(\Pi_{\mathbf{x}_t} \mathbf{c} \cdot \mathbf{x}_t)\}$ for $1 \sim T_1$ and $\ell_t^*(\mathbf{c}) = \max\{0, 1 - y_t(m_t^s \Pi_{\mathbf{x}_t^s} \mathbf{c} \cdot \mathbf{x}_t^s) - y_t(m_t^n \Pi_{\mathbf{x}_t^n} \mathbf{c} \cdot \mathbf{x}_t^n)\}$ for $T_1 \sim T_2$.

The proof of Lemma 4.1 can be found in Appendix A. Under the support of Lemma 4.1, we can go on to derive the following two theorems. Firstly, we derive the upper bound of our proposed algorithm under the ideal case, in which there is a perfect classifier $\mathbf{c} \in \mathbb{R}^{d_T}$ that can linearly separate over the sequence such that $\ell_t^*(\mathbf{c}) = 0$, i.e., for any $t \in [T]$, we obtain $y_t(\Pi_{\mathbf{x}_t} \mathbf{c} \cdot \mathbf{x}_t) \geq 1$ by projecting the classifier $\mathbf{c} \in \mathbb{R}^{d_T}$. Also, it implies the classifier $\mathbf{c} \in \mathbb{R}^{d_T}$ makes zero hinge loss during each iteration.

Theorem 1. Let $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_T, y_T)$ be a sequence of input instances, where $\mathbf{x}_t \in \mathbb{R}^{d_t}, d_t \leq d_{t+1}, y_t \in \{-1, +1\}$, and $\|\mathbf{x}_t\| \leq G$. Assume that there exists a classifier $\mathbf{c}$ such that $\ell_t^*(\mathbf{c}) = 0$ for all t . Then, the ideal number of errors made by the algorithm over T iterations is bounded by:

$$E[\sum_{t=1}^T R_t] \leq \rho \left(\frac{1+\nu}{2}\right)^2 \|\mathbf{c}\|^2 \quad (22)$$

where $\rho = \frac{1}{\nu} \min\{\frac{1}{C\gamma_t}, G^2\}$ and C is the aggressiveness parameter. Setting $\nu = 1$, we can obtain the upper bound as follows:

$$E[\sum_{t=1}^T R_t] \leq \rho \|\mathbf{c}\|^2. \quad (23)$$

More detailed derivation for Theorem 1 is provided in Appendix B. Since real-world datasets are usually not separable, we further generalize Theorem 1 and derive a linearly inseparable case, in which $\ell_t^*(\mathbf{c}) = 0$ may not always hold for all t . We present the expected mistake bound for the non-separable case and the derivative process of Theorem 2 can be found in Appendix C:

Theorem 2. Let $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_T, y_T)$ be a sequence of input instances, where $\mathbf{x}_t \in \mathbb{R}^{d_t}, d_t \leq d_{t+1}, y_t \in \{-1, +1\}$, and $\|\mathbf{x}_t\| \leq G$. Then, for any vector $\mathbf{c} \in \mathbb{R}^{d_T}$, the expected number of errors made by our algorithm is bounded by:

$$E[\sum_{t=1}^T R_t] \leq \rho \left(\frac{1+\nu}{2}\right)^2 \|\mathbf{c}\|^2 + \sum_{t=1}^T (\nu + 1) C \gamma_t \ell_t^*(\mathbf{c}) \quad (24)$$

where $\rho = \frac{1}{\nu} \min\{\frac{1}{C\gamma_t}, G^2\}$ and C is the aggressiveness parameter. Setting $\nu = 1$, we can obtain the upper bound as follows:

$$E[\sum_{t=1}^T R_t] \leq \rho \|\mathbf{c}\|^2 + \sum_{t=1}^T 2C \gamma_t \ell_t^*(\mathbf{c}). \quad (25)$$

5. Experiments

Firstly, the data sets and comparison algorithms with general settings are introduced. Then, we use the results of the following four experiments to illustrate the effectiveness of our algorithm.

5.1. Data sets and general settings

There are eleven synthetic data sets from UCI Repository² and LIBSVM³ Library, and three real-world data streams which are Twitter [47], RFID [48], weather [31] and covtypeNorm [49]. The Twitter dataset consists of real social media text data from the Twitter platform. Each sample typically represents a tweet, which can include user-generated text content, sentiment analysis labels, and more. User behavior and content vary over time, reflecting changes in topic popularity and shifts in sentiment trends. Hence, this dataset aligns well with our scenario of dynamic feature analysis. Hou et al. [48] collected the RFID data stream using RFID technology. Each RFID antenna continuously receives tag signals in every round. To ensure uninterrupted signal reception, new antennas are deployed alongside the existing ones before the old ones expire. This overlapping period enables data streams to be obtained from both historical and augmented feature spaces, demonstrating an increase in both the volume and dimensionality of the data over time. Thus, the RFID data collected by Hou et al. aligns with our assumptions. Weather was compiled from weather stations from more than 7000 weather stations around the world between 1949 and 1999. Generally, weather gradually changes with the seasons and human activities, so the setting can be viewed as a learning of dynamic data streams. The CovtypeNorm contains information about tree types and was collected by the United States Forest Service. The features in the dataset describe the environmental conditions of various regions, including different terrains, soils, and vegetation types. Because different sensors would have different lifespans, the collected features will undergo unpredictable feature evolution. Therefore, the CovtypeNorm data stream also satisfies our assumptions. The details of the 12 data sets used in our experiments are listed in Table 2. Also, for brevity, we fix the ratio T_p/T_n of positive and negative cases of all

Table 2
Brief description of data sets.

Data	Instance	Feature	Data	Instance	Feature
air	210	64	breast	277	9
cleve	296	13	heart	303	13
PIE ₃₂ × 32	340	1024	dna	949	180
msplice	600	240	svmguide	338	20
landsat	685	36	spambase	4601	57
twitter	4500	30	RFID	950	150
weather	10 000	8	covtypeNorm	10 000	54

datasets as 1/10 in training sets. The test set remains untouched and balanced.

The proposed algorithm has the advantage of simultaneously dealing with label scarcity, feature evolution, and data imbalance. To verify the validity of our algorithm in the above conclusion, we compare it with the following six proposed algorithms, such as CPA, PAUM, PEA, RPEA, FLLS, and RFLLS. PEA uses the Perceptron strategy for learning to deal with the label scarcity problem, and the current model is updated only if the model makes a mistake. CPA is only designed for the data imbalance problem, which uses a Cost-sensitive passive-aggressive algorithm based on prediction. FLLS is designed for incremental feature spaces with label scarcity, where both data volume and feature space simultaneously increase as a trapezoid. PAUM is also designed for imbalance data problems, it adds uneven margins based on Perceptron. RPEA is the random version of PEA, which queries labels uniformly and randomly. Similarly, RFLLS is the random version of FLLS. All the state-of-the-art algorithms are as follows:

- “PEA”: the Perceptron-based Active learning algorithm;
- “RPEA”: the Random PEA algorithm, which query labels uniformly and randomly;
- “CPA”: the Cost-sensitive Passive-aggressive Active learning algorithm;
- “PAUM”: the cost-sensitive Perceptron Algorithm with Uneven Margin;
- “FLLS”: the incremental Feature spaces Learning with Label Scarcity algorithm;
- “RFLLS”: the Random FLLS algorithm, which query labels uniformly and randomly;
- “OIFL”: the proposed algorithm;

All compared algorithms learn a linear classifier. Firstly, to ensure fairness, we select 20% of the instance as testing data for each dataset. Then we obtain 90% of the remaining negative cases randly and a certain number of positive cases as training data in the remaining data, which guarantees the proportion of minority and majority classes. The positive examples in training data and the testing data are constant in repeated experiments. We simulate feature evolution as follows, the training data is divided into three chunks, the first trunk carries 50% instances and features from S_1 . The second chunk contains 10% instances, which takes along all features with S_1 and S_2 . The last chunk includes 40% instances that S_2 gets inherited and S_1 loses 30% features randomly. The testing data contains all features.

Especially, for all the above algorithms, we set $\epsilon = 30$ and $B = 0.5$, which means there are 50% of features are used for iteration at each round t . We utilize Cross-validation for all the algorithms and datasets to obtain parameter C and σ , which are searched from $2^{[-5:5]}$. Parameter δ determines the query ratio, which is set as $2^{[-25:0.1:25]}$ for all datasets. We used common measures of online learning, such as ACC, AUC, F-measure, and G-mean, obtained from test data to evaluate online classification performance after the T rounds. For each data set, we run the experiment 20 times and report an average performance.

5.2. Evaluation on fixed ratio of queries

In this subsection, the first experiment is to evaluate the performance of these algorithms with the fixed proportion of queries, where we present the experimental results of seven algorithms on nine benchmark datasets. Tables 3 and 4 show the average performance indicators(ACC values, AUC values, F-measure, and G-mean) and runtime of these seven methods at different query rates of about 10% and 20%, respectively. We highlight the best performance results and their comparability results in boldface, through a paired t-test at the 95% significance level.

Several conclusions can be acquired from the results. First of all, when querying the same proportion of labeled data, the performance indexes of CPA and PAUM are always better than that of other algorithms except OIFL in most datasets, which implies the importance of considering class imbalance in classification tasks. Meanwhile, the effectiveness of the cost-based reweighting strategy is verified in the case of solving the imbalanced data streams. Second, it is clear that our proposed algorithm significantly outperforms other comparison methods in all performance metrics when querying the same ratio of labels, which means that the proposed strategy can enrich the information of the minority class in the query instance and can actively adapt to the unpredictable feature evolution process. Third, it can be observed that both PEA and FLLS are superior to corresponding random versions in any performance indicator result, which verifies the effectiveness of the label query strategy that can actively select those fairly informative instances to query labels. Finally, we notice that the running time of OIFL is always higher than other algorithms, the main reason is that to preserve the feature information and maintain the feature weights, we recalculate the variance for each instance that enters the model. By comprehensively comparing the results of our OIFL algorithm on the same dataset under different query rates, we can observe that sometimes the algorithm achieves better results at lower query rates, which vividly implies the effectiveness of our active learning strategy and adaptive weighting strategy. In addition, as can be seen from the numerical results in Tables 3 and 4, for practical applications, the extra computational costs are not very large and are affordable.

5.3. Evaluation on varied ratios of queries

This experiment is to evaluate the performance of the proposed algorithm by changing the ratio of queries to compare different advanced algorithms. Fig. 2 shows the average AUC values for different query ratios. The AUC is not sensitive to whether the positive and negative samples are balanced and can make a reasonable assessment in any case.

From the experimental results can be concluded that the following observations. First of all, our proposed algorithm usually shows a surge in AUC values in the early stage as the query rate increases, which shows that the proposed algorithm can respond quickly to the selected case and make a good response to the uncertainties carried by the instance. Second, It can be noted that the AUC value curve of other comparison algorithms changes more smoothly with the increase in query rate, and sometimes shows a decline. Meanwhile, we find that our proposed algorithm OIFL levels off rapidly and much better than the others. The interesting observation indicates the effectiveness of the proposed algorithms in alleviating the problem of unpredictable feature evolution with imbalanced data streams. Finally, when the query ratio is 1 (equivalent to the query the label for any instance), OIFL achieves significantly higher sum performance, suggesting that the biased penalty function does effectively optimize the performance measures.

² <http://archive.ics.uci.edu/ml>.

³ <http://www.csie.ntu.edu.tw/~cjlin/libsvm>.

Table 3Experimental results (mean $\pm$ std(rank)) on the twelve datasets with the query ratio fixed to about 10%, and win/tie/loss counts of OIFL versus the other methods.

Data	Methods	Measures						Win/tie/loss
		ACC	AUC	F-measure	G-mean	Times	Query (%)	
air	PEA	.845 $\pm$.039	.879 $\pm$.034	.831 $\pm$.049	.839 $\pm$.043	.0014	10.31 $\pm$ 4.34	24/0/0
	RPEA	.830 $\pm$.060	.844 $\pm$.076	.818 $\pm$.065	.826 $\pm$.062	.0003	10.00 $\pm$ 0.00	
	FLLS	.850 $\pm$.038	.889 $\pm$.037	.837 $\pm$.045	.846 $\pm$.040	.0014	10.19 $\pm$ 3.02	
	RFLLS	.829 $\pm$.049	.844 $\pm$.064	.815 $\pm$.067	.824 $\pm$.056	.0003	10.00 $\pm$ 0.00	
	CPA	.898 $\pm$.030	.933 $\pm$.028	.891 $\pm$.032	.894 $\pm$.013	.0013	9.69 $\pm$ 2.05	
	PAUM	.829 $\pm$.042	.858 $\pm$.047	.821 $\pm$.040	.824 $\pm$.042	.0013	9.88 $\pm$ 2.35	
	OIFL	.919 $\pm$.024	.954 $\pm$.023	.914 $\pm$.027	.917 $\pm$.025	.0018	10.06 $\pm$ 3.38	
breast	PEA	.651 $\pm$.027	.652 $\pm$.026	.644 $\pm$.048	.646 $\pm$.031	.0019	9.55 $\pm$ 3.33	24/0/0
	RPEA	.639 $\pm$.054	.633 $\pm$.055	.648 $\pm$.057	.637 $\pm$.054	.0004	10.00 $\pm$ 0.00	
	FLLS	.642 $\pm$.029	.638 $\pm$.029	.648 $\pm$.044	.639 $\pm$.029	.0019	10.10 $\pm$ 2.51	
	RFLLS	.616 $\pm$.060	.633 $\pm$.061	.615 $\pm$.069	.612 $\pm$.060	.0004	10.00 $\pm$ 0.00	
	CPA	.662 $\pm$.007	.657 $\pm$.008	.671 $\pm$.011	.660 $\pm$.009	.0019	9.84 $\pm$ 2.98	
	PAUM	.621 $\pm$.024	.623 $\pm$.026	.595 $\pm$.061	.608 $\pm$.032	.0018	10.32 $\pm$ 1.93	
	OIFL	.704 $\pm$.021	.710 $\pm$.022	.692 $\pm$.035	.700 $\pm$.024	.0034	9.81 $\pm$ 3.78	
cleve	PEA	.740 $\pm$.033	.780 $\pm$.038	.741 $\pm$.035	.738 $\pm$.033	.0019	9.86 $\pm$ 3.24	24/0/0
	RPEA	.706 $\pm$.056	.738 $\pm$.058	.706 $\pm$.065	.703 $\pm$.056	.0003	10.00 $\pm$ 0.00	
	FLLS	.741 $\pm$.040	.782 $\pm$.038	.741 $\pm$.042	.738 $\pm$.040	.0019	9.95 $\pm$ 2.63	
	RFLLS	.720 $\pm$.071	.755 $\pm$.070	.716 $\pm$.072	.718 $\pm$.072	.0003	10.00 $\pm$ 0.00	
	CPA	.760 $\pm$.018	.786 $\pm$.014	.757 $\pm$.017	.758 $\pm$.018	.0019	10.42 $\pm$ 3.08	
	PAUM	.686 $\pm$.045	.727 $\pm$.037	.671 $\pm$.048	.683 $\pm$.045	.0020	10.28 $\pm$ 1.98	
	OIFL	.804 $\pm$.030	.844 $\pm$.029	.804 $\pm$.032	.802 $\pm$.030	.0023	9.95 $\pm$ 2.41	
heart	PEA	.780 $\pm$.046	.799 $\pm$.048	.790 $\pm$.045	.783 $\pm$.046	.0016	10.39 $\pm$ 2.67	24/0/0
	RPEA	.752 $\pm$.064	.764 $\pm$.059	.766 $\pm$.054	.747 $\pm$.068	.0003	10.00 $\pm$ 0.00	
	FLLS	.770 $\pm$.043	.774 $\pm$.047	.780 $\pm$.039	.737 $\pm$.045	.0015	10.12 $\pm$ 3.64	
	RFLLS	.713 $\pm$.092	.724 $\pm$.094	.732 $\pm$.083	.707 $\pm$.096	.0003	10.00 $\pm$ 0.00	
	CPA	.693 $\pm$.019	.721 $\pm$.021	.711 $\pm$.021	.693 $\pm$.022	.0015	10.19 $\pm$ 4.44	
	PAUM	.660 $\pm$.023	.677 $\pm$.025	.681 $\pm$.029	.652 $\pm$.028	.0015	10.08 $\pm$ 2.00	
	OIFL	.823 $\pm$.036	.850 $\pm$.034	.826 $\pm$.022	.822 $\pm$.036	.0028	9.96 $\pm$ 3.47	
PIE_32 $\times$ 32	PEA	.692 $\pm$.066	.707 $\pm$.050	.699 $\pm$.082	.683 $\pm$.061	.0055	9.63 $\pm$ 3.38	24/0/0
	RPEA	.655 $\pm$.044	.662 $\pm$.048	.671 $\pm$.042	.651 $\pm$.046	.0020	10.00 $\pm$ 0.00	
	FLLS	.710 $\pm$.081	.715 $\pm$.067	.723 $\pm$.083	.705 $\pm$.078	.0055	10.19 $\pm$ 2.75	
	RFLLS	.643 $\pm$.073	.658 $\pm$.064	.662 $\pm$.070	.635 $\pm$.074	.0027	10.00 $\pm$ 0.00	
	CPA	.834 $\pm$.081	.851 $\pm$.075	.840 $\pm$.077	.830 $\pm$.081	.0043	9.70 $\pm$ 2.23	
	PAUM	.745 $\pm$.063	.780 $\pm$.065	.751 $\pm$.068	.739 $\pm$.061	.0043	9.70 $\pm$ 1.65	
	OIFL	.974 $\pm$.032	.960 $\pm$.024	.973 $\pm$.033	.973 $\pm$.032	.0057	9.85 $\pm$ 2.44	
dna	PEA	.754 $\pm$.042	.770 $\pm$.039	.755 $\pm$.042	.753 $\pm$.042	.0090	10.05 $\pm$ 1.75	24/0/0
	RPEA	.720 $\pm$.066	.736 $\pm$.062	.723 $\pm$.069	.718 $\pm$.066	.0020	10.00 $\pm$ 0.00	
	FLLS	.727 $\pm$.038	.745 $\pm$.035	.726 $\pm$.040	.727 $\pm$.039	.0020	10.50 $\pm$ 1.38	
	RFLLS	.695 $\pm$.051	.709 $\pm$.049	.699 $\pm$.048	.694 $\pm$.052	.0073	10.00 $\pm$ 0.00	
	CPA	.729 $\pm$.036	.742 $\pm$.033	.723 $\pm$.028	.727 $\pm$.035	.0066	10.01 $\pm$ 1.20	
	PAUM	.671 $\pm$.046	.686 $\pm$.045	.664 $\pm$.045	.669 $\pm$.045	.0083	10.18 $\pm$ 1.33	
	OIFL	.845 $\pm$.021	.859 $\pm$.021	.842 $\pm$.021	.844 $\pm$.021	.0184	10.22 $\pm$ 1.89	
msplice	PEA	.721 $\pm$.053	.708 $\pm$.053	.723 $\pm$.052	.720 $\pm$.053	.0049	10.34 $\pm$ 1.92	24/0/0
	RPEA	.675 $\pm$.041	.660 $\pm$.045	.684 $\pm$.042	.674 $\pm$.042	.0018	10.00 $\pm$ 0.00	
	FLLS	.727 $\pm$.046	.711 $\pm$.049	.731 $\pm$.049	.725 $\pm$.046	.0050	10.27 $\pm$ 2.00	
	RFLLS	.669 $\pm$.063	.654 $\pm$.067	.676 $\pm$.061	.667 $\pm$.064	.0019	10.00 $\pm$ 0.00	
	CPA	.847 $\pm$.030	.843 $\pm$.030	.844 $\pm$.027	.846 $\pm$.030	.0045	10.25 $\pm$ 1.97	
	PAUM	.781 $\pm$.055	.772 $\pm$.060	.780 $\pm$.053	.780 $\pm$.055	.0042	10.13 $\pm$ 1.42	
	OIFL	.884 $\pm$.020	.884 $\pm$.020	.882 $\pm$.022	.883 $\pm$.021	.0057	9.77 $\pm$ 2.05	
svmguid	PEA	.789 $\pm$.055	.842 $\pm$.046	.773 $\pm$.059	.784 $\pm$.055	.0011	9.73 $\pm$ 3.00	24/0/0
	RPEA	.754 $\pm$.083	.804 $\pm$.070	.737 $\pm$.095	.750 $\pm$.086	.0002	10.00 $\pm$ 0.00	
	FLLS	.798 $\pm$.042	.847 $\pm$.043	.777 $\pm$.045	.791 $\pm$.043	.0011	10.33 $\pm$ 2.83	
	RFLLS	.737 $\pm$.060	.789 $\pm$.054	.715 $\pm$.059	.730 $\pm$.058	.0002	10.00 $\pm$ 0.00	
	CPA	.796 $\pm$.037	.853 $\pm$.023	.763 $\pm$.050	.783 $\pm$.043	.0011	10.00 $\pm$ 3.32	
	PAUM	.732 $\pm$.037	.762 $\pm$.030	.707 $\pm$.043	.725 $\pm$.038	.0011	9.78 $\pm$ 2.11	
	OIFL	.842 $\pm$.034	.889 $\pm$.016	.824 $\pm$.041	.845 $\pm$.037	.0014	10.22 $\pm$ 4.72	
landsate	PEA	.996 $\pm$.004	.998 $\pm$.002	.996 $\pm$.004	.996 $\pm$.004	.0031	10.06 $\pm$ 3.40	20/4/0
	RPEA	.960 $\pm$.030	.970 $\pm$.025	.963 $\pm$.028	.959 $\pm$.032	.0004	10.00 $\pm$ 0.00	
	FLLS	.972 $\pm$.023	.982 $\pm$.020	.973 $\pm$.021	.971 $\pm$.023	.0032	9.55 $\pm$ 1.19	
	RFLLS	.934 $\pm$.024	.950 $\pm$.020	.939 $\pm$.021	.931 $\pm$.026	.0005	10.00 $\pm$ 0.00	
	CPA	.964 $\pm$.028	.972 $\pm$.023	.965 $\pm$.026	.963 $\pm$.029	.0031	10.06 $\pm$ 2.65	
	PAUM	.931 $\pm$.022	.946 $\pm$.015	.935 $\pm$.020	.928 $\pm$.023	.0031	10.11 $\pm$ 1.79	
	OIFL	.998 $\pm$.002	.999 $\pm$.001	.998 $\pm$.002	.998 $\pm$.002	.0039	9.78 $\pm$ 2.22	
spambase	PEA	.677 $\pm$.052	.660 $\pm$.051	.710 $\pm$.053	.647 $\pm$.073	.0122	10.16 $\pm$ 1.33	
	RPEA	.672 $\pm$.036	.656 $\pm$.036	.689 $\pm$.064	.658 $\pm$.037	.0024	10.00 $\pm$ 0.00	
	FLLS	.673 $\pm$.048	.658 $\pm$.046	.690 $\pm$.048	.643 $\pm$.059	.0131	10.09 $\pm$ 0.70	
	RFLLS	.588 $\pm$.069	.571 $\pm$.070	.555 $\pm$.122	.564 $\pm$.078	.0035	10.00 $\pm$ 0.00	

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Table 3 (continued).

Data	Methods	Measures						Win/tie/loss
		ACC	AUC	F-measure	G-mean	Times	Query (%)	
twitter	CPA	.734 $\pm$.024	.703 $\pm$.024	.731 $\pm$.054	.685 $\pm$.040	.0127	10.07 $\pm$ 0.90	24/0/0
	PAUM	.651 $\pm$.021	.639 $\pm$.022	.683 $\pm$.084	.605 $\pm$.028	.0118	10.69 $\pm$ 1.48	
	OIFL	.763 $\pm$.020	.754 $\pm$.020	.774 $\pm$.033	.757 $\pm$.029	.0189	10.15 $\pm$ 0.76	
	PEA	.630 $\pm$.017	.630 $\pm$.017	.632 $\pm$.023	.629 $\pm$.016	.0145	10.03 $\pm$ 0.49	24/0/0
	RPEA	.582 $\pm$.070	.580 $\pm$.071	.587 $\pm$.059	.579 $\pm$.072	.0028	10.00 $\pm$ 0.00	
	FLLS	.617 $\pm$.040	.616 $\pm$.039	.622 $\pm$.042	.616 $\pm$.039	.0154	9.43 $\pm$ 0.41	
	RFLLS	.567 $\pm$.077	.566 $\pm$.076	.573 $\pm$.108	.565 $\pm$.077	.0037	10.00 $\pm$ 0.00	
	CPA	.643 $\pm$.011	.642 $\pm$.008	.653 $\pm$.007	.641 $\pm$.009	.0141	10.13 $\pm$ 0.36	
RFID	PAUM	.612 $\pm$.023	.610 $\pm$.023	.618 $\pm$.019	.610 $\pm$.024	.0143	9.63 $\pm$ 0.65	24/0/0
	OIFL	.658 $\pm$.008	.657 $\pm$.008	.653 $\pm$.014	.657 $\pm$.008	.0180	10.18 $\pm$ 0.46	
	PEA	.740 $\pm$.048	.777 $\pm$.046	.734 $\pm$.047	.724 $\pm$.056	.0014	9.96 $\pm$ 1.47	24/0/0
	RPEA	.656 $\pm$.143	.692 $\pm$.132	.680 $\pm$.096	.615 $\pm$.182	.0004	10.00 $\pm$ 0.00	
	FLLS	.729 $\pm$.047	.707 $\pm$.051	.707 $\pm$.071	.712 $\pm$.058	.0015	9.64 $\pm$ 2.01	
	RFLLS	.642 $\pm$.118	.685 $\pm$.111	.674 $\pm$.085	.570 $\pm$.170	.0005	10.00 $\pm$ 0.00	
	CPA	.886 $\pm$.026	.886 $\pm$.030	.862 $\pm$.032	.864 $\pm$.026	.0014	10.00 $\pm$ 2.23	
	PAUM	.819 $\pm$.056	.833 $\pm$.049	.797 $\pm$.080	.809 $\pm$.067	.0012	9.78 $\pm$ 1.91	
	OIFL	.907 $\pm$.028	.918 $\pm$.027	.903 $\pm$.033	.905 $\pm$.029	.0042	9.78 $\pm$ 1.83	

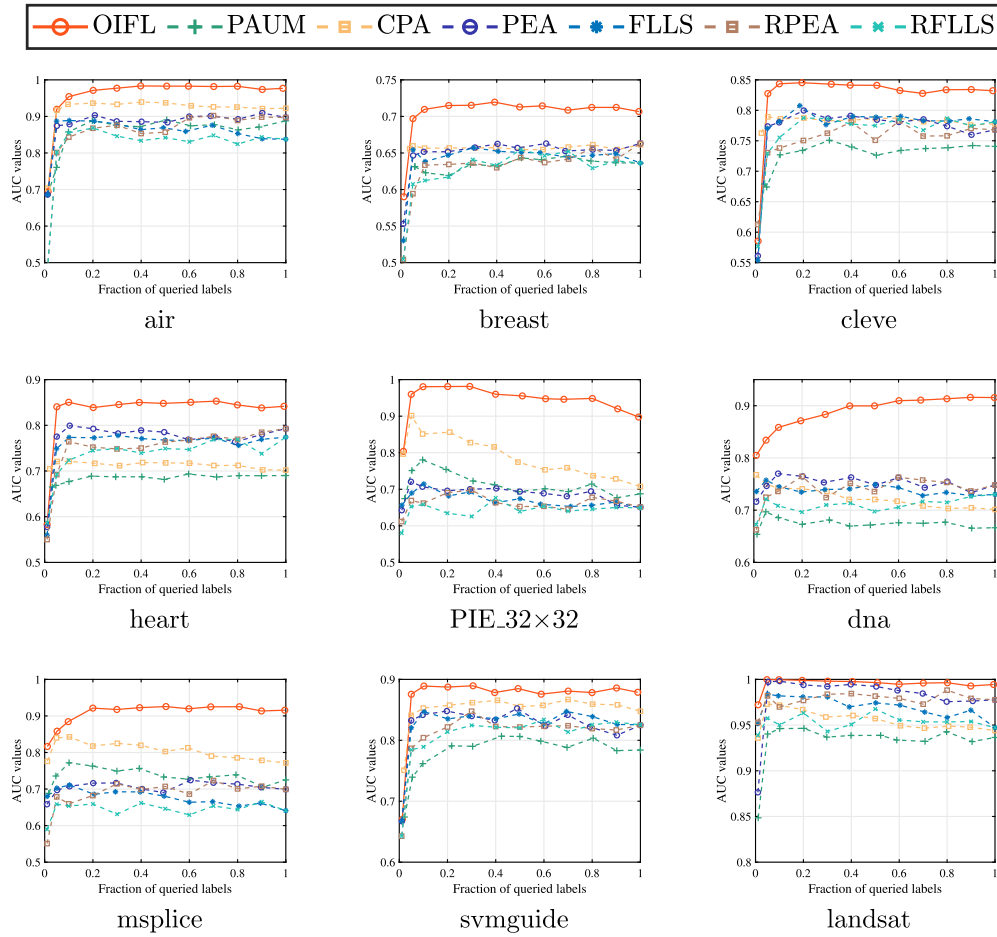


Fig. 2. Experimental AUC results on the real-world datasets: air, breast, cleve, heart, PIE₃₂ × 32, dna, msplc, svmguide and landsat. The curve shows the average learning AUC results over the fraction of queried labels.

5.4. Analysis of sparsity strategy

In this subsection, we test the effectiveness of model sparsity in our algorithm through experiments. We conduct experiences on six datasets (air, cleve, dna, heart, msplc, and svmguide) under different values of parameter B, which represents the proportion of selected features. Also, to make the experience more convincing, we adjust for different query studios (10%, 50%, and 90%) in the experiment and

tune B in {0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9}. We use the stacked bar to show the results in Fig. 3.

From these results shown in the above figure, we can draw the following conclusions. First, our algorithm achieves better performance at the same B settings for different data sets. The results show that the relative uncertainty of features can be used to avoid bias towards features. Then, Our algorithm achieves relatively stable performance under the same parameter B setting with different query studios on most datasets, which indicates that the sparse strategy of OIFL can

Table 4

Experimental results (mean $\pm$ std(rank)) on the twelve datasets with the query ratio fixed to about 20%, and win/tie/loss counts of OIFL versus the other methods.

Data	Methods	Measures						Win/tie/loss
		ACC	AUC	F-measure	G-mean	Times	Query (%)	
air	PEA	.876 $\pm$.050	.903 $\pm$.053	.874 $\pm$.049	.875 $\pm$.050	.0017	20.08 $\pm$ 4.50	23/0/1
	RPEA	.838 $\pm$.046	.868 $\pm$.040	.828 $\pm$.051	.834 $\pm$.048	.0005	20.00 $\pm$ 0.00	
	FLLS	.860 $\pm$.029	.887 $\pm$.035	.852 $\pm$.032	.857 $\pm$.029	.0019	20.62 $\pm$ 4.52	
	RFLLS	.841 $\pm$.058	.868 $\pm$.063	.833 $\pm$.062	.837 $\pm$.060	.0005	20.00 $\pm$ 0.00	
	CPA	.986 $\pm$.026	.936 $\pm$.023	.891 $\pm$.029	.895 $\pm$.027	.0017	20.00 $\pm$ 3.75	
	PAUM	.856 $\pm$.046	.889 $\pm$.039	.845 $\pm$.051	.852 $\pm$.047	.0014	19.75 $\pm$ 3.41	
	OIFL	.944 $\pm$.032	.971 $\pm$.021	.941 $\pm$.035	.942 $\pm$.033	.0020	20.12 $\pm$ 5.39	
breast	PEA	.650 $\pm$.020	.652 $\pm$.022	.633 $\pm$.044	.642 $\pm$.024	.0020	19.61 $\pm$ 3.88	24/0/0
	RPEA	.640 $\pm$.043	.634 $\pm$.046	.636 $\pm$.056	.637 $\pm$.046	.0005	10.00 $\pm$ 0.00	
	FLLS	.650 $\pm$.036	.647 $\pm$.036	.654 $\pm$.048	.646 $\pm$.038	.0020	19.68 $\pm$ 2.99	
	RFLLS	.620 $\pm$.068	.634 $\pm$.068	.628 $\pm$.074	.615 $\pm$.073	.0005	20.00 $\pm$ 0.00	
	CPA	.661 $\pm$.016	.657 $\pm$.016	.661 $\pm$.028	.657 $\pm$.017	.0019	20.16 $\pm$ 4.67	
	PAUM	.614 $\pm$.028	.619 $\pm$.032	.573 $\pm$.054	.597 $\pm$.032	.0020	20.00 $\pm$ 4.33	
	OIFL	.711 $\pm$.020	.715 $\pm$.019	.696 $\pm$.032	.708 $\pm$.019	.0036	19.97 $\pm$ 4.81	
cleve	PEA	.757 $\pm$.036	.800 $\pm$.037	.755 $\pm$.042	.755 $\pm$.037	.0019	20.14 $\pm$ 4.16	24/0/0
	RPEA	.717 $\pm$.050	.750 $\pm$.050	.705 $\pm$.061	.714 $\pm$.052	.0004	10.00 $\pm$ 0.00	
	FLLS	.770 $\pm$.041	.808 $\pm$.040	.764 $\pm$.050	.768 $\pm$.043	.0020	18.50 $\pm$ 3.38	
	RFLLS	.746 $\pm$.043	.789 $\pm$.043	.744 $\pm$.043	.744 $\pm$.063	.0004	20.00 $\pm$ 0.00	
	CPA	.759 $\pm$.016	.787 $\pm$.013	.756 $\pm$.016	.758 $\pm$.016	.0020	20.19 $\pm$ 3.42	
	PAUM	.708 $\pm$.024	.737 $\pm$.023	.699 $\pm$.031	.706 $\pm$.024	.0019	19.63 $\pm$ 3.89	
	OIFL	.807 $\pm$.030	.845 $\pm$.032	.807 $\pm$.029	.806 $\pm$.030	.0027	19.39 $\pm$ 3.87	
heart	PEA	.780 $\pm$.040	.793 $\pm$.046	.786 $\pm$.038	.778 $\pm$.040	.0015	20.16 $\pm$ 5.15	24/0/0
	RPEA	.739 $\pm$.057	.752 $\pm$.059	.740 $\pm$.069	.738 $\pm$.057	.0004	20.00 $\pm$ 0.00	
	FLLS	.765 $\pm$.046	.773 $\pm$.046	.772 $\pm$.045	.763 $\pm$.045	.0016	20.54 $\pm$ 5.15	
	RFLLS	.729 $\pm$.057	.745 $\pm$.056	.744 $\pm$.050	.724 $\pm$.061	.0004	20.00 $\pm$ 0.00	
	CPA	.686 $\pm$.014	.717 $\pm$.012	.714 $\pm$.019	.686 $\pm$.016	.0016	20.81 $\pm$ 3.91	
	PAUM	.667 $\pm$.018	.689 $\pm$.017	.699 $\pm$.024	.655 $\pm$.022	.0016	19.38 $\pm$ 3.95	
	OIFL	.831 $\pm$.032	.839 $\pm$.030	.833 $\pm$.031	.831 $\pm$.032	.0026	20.16 $\pm$ 3.32	
PIE_32 $\times$ 32	PEA	.684 $\pm$.058	.696 $\pm$.052	.683 $\pm$.072	.680 $\pm$.055	.0044	19.93 $\pm$ 2.29	24/0/0
	RPEA	.675 $\pm$.061	.691 $\pm$.063	.691 $\pm$.064	.669 $\pm$.059	.0027	20.00 $\pm$ 0.00	
	FLLS	.672 $\pm$.066	.682 $\pm$.058	.693 $\pm$.069	.663 $\pm$.064	.0003	20.49 $\pm$ 3.60	
	RFLLS	.634 $\pm$.056	.635 $\pm$.042	.651 $\pm$.059	.628 $\pm$.057	.0038	20.00 $\pm$ 0.00	
	CPA	.842 $\pm$.083	.856 $\pm$.083	.849 $\pm$.081	.836 $\pm$.085	.0048	20.93 $\pm$ 3.40	
	PAUM	.740 $\pm$.089	.754 $\pm$.072	.748 $\pm$.093	.731 $\pm$.089	.0037	19.40 $\pm$ 3.21	
	OIFL	.972 $\pm$.031	.981 $\pm$.022	.973 $\pm$.030	.972 $\pm$.032	.0067	20.07 $\pm$ 3.37	
dna	PEA	.749 $\pm$.047	.765 $\pm$.043	.746 $\pm$.050	.748 $\pm$.047	.0084	20.25 $\pm$ 2.28	24/0/0
	RPEA	.748 $\pm$.046	.765 $\pm$.040	.753 $\pm$.046	.748 $\pm$.046	.0018	20.00 $\pm$ 0.00	
	FLLS	.713 $\pm$.033	.735 $\pm$.032	.719 $\pm$.038	.711 $\pm$.032	.0100	19.67 $\pm$ 1.71	
	RFLLS	.679 $\pm$.064	.696 $\pm$.064	.684 $\pm$.063	.678 $\pm$.064	.0030	20.00 $\pm$ 0.00	
	CPA	.728 $\pm$.042	.741 $\pm$.040	.728 $\pm$.039	.727 $\pm$.042	.0073	20.04 $\pm$ 1.91	
	PAUM	.658 $\pm$.041	.673 $\pm$.038	.657 $\pm$.038	.657 $\pm$.041	.0089	20.10 $\pm$ 2.06	
	OIFL	.860 $\pm$.015	.871 $\pm$.018	.856 $\pm$.016	.859 $\pm$.015	.0190	19.44 $\pm$ 2.01	
msplice	PEA	.732 $\pm$.045	.716 $\pm$.046	.732 $\pm$.041	.731 $\pm$.045	.0059	20.13 $\pm$ 2.15	24/0/0
	RPEA	.702 $\pm$.063	.682 $\pm$.062	.704 $\pm$.060	.701 $\pm$.064	.0023	20.00 $\pm$ 0.00	
	FLLS	.699 $\pm$.055	.686 $\pm$.058	.707 $\pm$.057	.696 $\pm$.054	.0070	20.55 $\pm$ 2.03	
	RFLLS	.676 $\pm$.060	.659 $\pm$.064	.686 $\pm$.058	.675 $\pm$.060	.0025	20.00 $\pm$ 0.00	
	CPA	.820 $\pm$.035	.818 $\pm$.039	.820 $\pm$.034	.818 $\pm$.037	.0059	19.96 $\pm$ 2.77	
	PAUM	.770 $\pm$.053	.763 $\pm$.058	.773 $\pm$.049	.769 $\pm$.053	.0059	19.83 $\pm$ 1.94	
	OIFL	.923 $\pm$.021	.921 $\pm$.023	.922 $\pm$.021	.992 $\pm$.021	.0083	19.83 $\pm$ 3.18	
svmguide	PEA	.797 $\pm$.039	.849 $\pm$.026	.771 $\pm$.052	.788 $\pm$.044	.0011	19.73 $\pm$ 4.22	24/0/0
	RPEA	.772 $\pm$.048	.822 $\pm$.046	.747 $\pm$.056	.763 $\pm$.050	.0003	20.00 $\pm$ 0.00	
	FLLS	.791 $\pm$.050	.835 $\pm$.032	.769 $\pm$.065	.784 $\pm$.055	.0011	20.16 $\pm$ 4.07	
	RFLLS	.754 $\pm$.069	.814 $\pm$.065	.729 $\pm$.072	.746 $\pm$.069	.0003	20.00 $\pm$ 0.00	
	CPA	.809 $\pm$.025	.858 $\pm$.017	.781 $\pm$.053	.798 $\pm$.030	.0012	20.87 $\pm$ 3.59	
	PAUM	.737 $\pm$.037	.791 $\pm$.029	.712 $\pm$.036	.742 $\pm$.037	.0011	21.74 $\pm$ 2.77	
	OIFL	.842 $\pm$.030	.887 $\pm$.014	.822 $\pm$.040	.834 $\pm$.034	.0016	20.16 $\pm$ 5.33	
landsate	PEA	.988 $\pm$.012	.994 $\pm$.006	.988 $\pm$.012	.988 $\pm$.012	.0032	20.03 $\pm$ 5.15	23/1/0
	RPEA	.966 $\pm$.033	.977 $\pm$.023	.968 $\pm$.030	.965 $\pm$.032	.0004	20.00 $\pm$ 0.00	
	FLLS	.970 $\pm$.023	.981 $\pm$.019	.971 $\pm$.022	.969 $\pm$.024	.0035	19.80 $\pm$ 4.52	
	RFLLS	.949 $\pm$.031	.963 $\pm$.026	.952 $\pm$.027	.947 $\pm$.034	.0007	20.00 $\pm$ 0.00	
	CPA	.957 $\pm$.026	.967 $\pm$.023	.959 $\pm$.025	.956 $\pm$.027	.0032	19.69 $\pm$ 3.42	
	PAUM	.933 $\pm$.022	.946 $\pm$.016	.937 $\pm$.019	.930 $\pm$.023	.0033	20.22 $\pm$ 3.22	
	OIFL	.996 $\pm$.004	.999 $\pm$.001	.996 $\pm$.004	.996 $\pm$.004	.0053	19.49 $\pm$ 3.31	
spambase	PEA	.691 $\pm$.046	.677 $\pm$.046	.709 $\pm$.071	.666 $\pm$.054	.0123	19.86 $\pm$ 2.90	
	RPEA	.672 $\pm$.042	.657 $\pm$.043	.687 $\pm$.079	.642 $\pm$.056	.0027	20.00 $\pm$ 0.00	
	FLLS	.626 $\pm$.049	.609 $\pm$.047	.680 $\pm$.067	.574 $\pm$.066	.0142	19.73 $\pm$ 1.45	
	RFLLS	.570 $\pm$.043	.554 $\pm$.044	.509 $\pm$.079	.554 $\pm$.053	.0046	20.00 $\pm$ 0.00	

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Table 4 (continued).

Data	Methods	Measures						Win/tie/loss
		ACC	AUC	F-measure	G-mean	Times	Query (%)	
twitter	CPA	.715 $\pm$.040	.700 $\pm$.038	.712 $\pm$.074	.700 $\pm$.042	.0132	20.23 $\pm$ 1.76	24/0/0
	PAUM	.662 $\pm$.035	.654 $\pm$.036	.694 $\pm$.069	.620 $\pm$.043	.0134	19.94 $\pm$ 1.88	
	OIFL	.758 $\pm$.029	.747 $\pm$.031	.774 $\pm$.028	.751 $\pm$.035	.0173	20.58 $\pm$ 0.84	
	PEA	.631 $\pm$.017	.630 $\pm$.017	.638 $\pm$.014	.631 $\pm$.017	.0137	20.05 $\pm$ 0.60	24/0/0
	RPEA	.611 $\pm$.022	.610 $\pm$.022	.614 $\pm$.027	.609 $\pm$.023	.0030	20.00 $\pm$ 0.00	
	FLLS	.615 $\pm$.033	.614 $\pm$.034	.620 $\pm$.034	.614 $\pm$.033	.0157	20.01 $\pm$ 0.71	
	RFLLS	.587 $\pm$.073	.586 $\pm$.073	.595 $\pm$.089	.583 $\pm$.073	.0049	20.00 $\pm$ 0.00	
	CPA	.634 $\pm$.008	.633 $\pm$.008	.640 $\pm$.008	.640 $\pm$.008	.0149	19.84 $\pm$ 0.53	
RFID	PAUM	.611 $\pm$.021	.610 $\pm$.021	.612 $\pm$.022	.610 $\pm$.021	.0143	20.13 $\pm$ 0.84	24/0/0
	OIFL	.660 $\pm$.007	.659 $\pm$.007	.663 $\pm$.009	.660 $\pm$.008	.0183	19.88 $\pm$ 1.04	
	PEA	.751 $\pm$.027	.777 $\pm$.028	.716 $\pm$.045	.733 $\pm$.032	.0015	19.78 $\pm$ 3.31	24/0/0
	RPEA	.657 $\pm$.076	.690 $\pm$.072	.608 $\pm$.139	.624 $\pm$.091	.0004	20.00 $\pm$ 0.00	
	FLLS	.729 $\pm$.053	.754 $\pm$.043	.713 $\pm$.072	.709 $\pm$.063	.0019	20.27 $\pm$ 1.96	
	RFLLS	.600 $\pm$.108	.645 $\pm$.091	.659 $\pm$.076	.536 $\pm$.148	.0005	20.00 $\pm$ 0.00	
	CPA	.857 $\pm$.024	.877 $\pm$.038	.849 $\pm$.042	.853 $\pm$.040	.0014	20.40 $\pm$ 2.54	
	PAUM	.827 $\pm$.060	.849 $\pm$.055	.809 $\pm$.068	.819 $\pm$.065	.0012	20.44 $\pm$ 3.33	
	OIFL	.909 $\pm$.024	.923 $\pm$.017	.908 $\pm$.024	.908 $\pm$.023	.0041	20.04 $\pm$ 3.90	

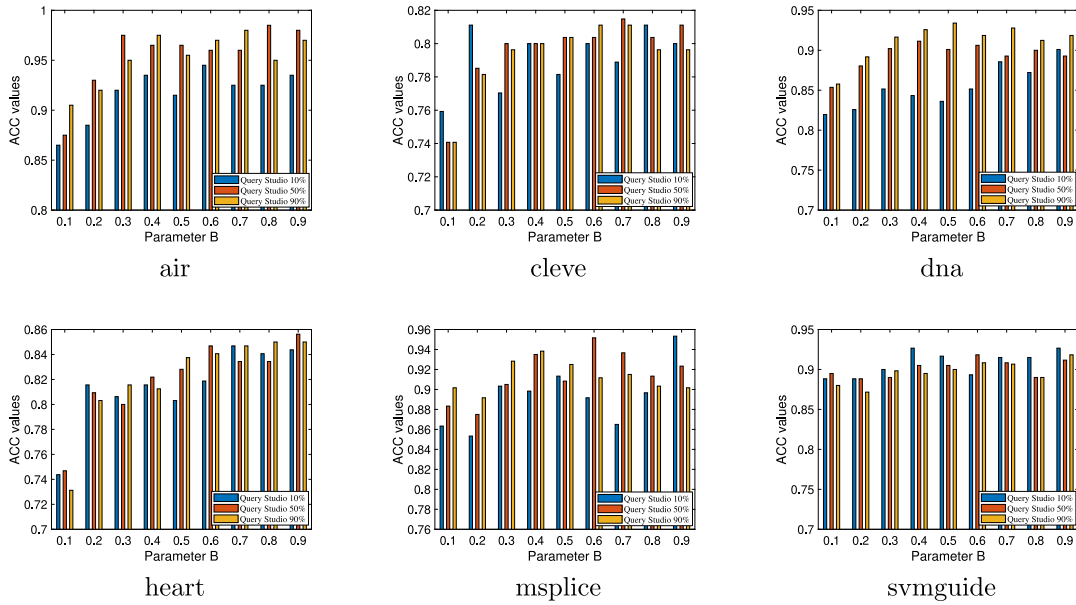


Fig. 3. Experimental ACC performance of OIFL in different query ratios(10%, 50%, 90%) of six datasets(air, cleve, dna, heart, msplce, and svmguide) under different values of parameter B.

effectively truncate unimportant features as well as avoid feature bias when training the classifier. Last but not least, the ACC values decrease instead of increasing as parameter B increases on most datasets in different query studios. We can conclude by comparison that larger B values do not necessarily lead to better performance, which suggests that sparse strategies can not only improve the efficiency of memory utilization and running time but also ensure that the proposed algorithm has superior performance.

5.5. Parameter analysis

As described in Section 3, the classification performance of OIFL is related to two parameters: penalty cost parameter C and scale factor σ . To further analyze the effects of two parameters on the performance of our model, we conducted ablation studies on four representative datasets, i.e. air, cleve, heart, and msplce. we fix B = 0.5 and tune C and σ in $2^{[-5:1:5]}$. All of these experiments below were conducted with a fixed query rate of around 50%.

The parameter C enables the model to achieve better generalization performance on the dataset by adjusting the slack variable ξ . Scale

factor σ adjusts the cost and convergence speed of the current class to obtain a better classifier. As shown in Fig. 4, it shows F-measure results of OIFL in different values of C and σ . From these results, we can find an interesting phenomenon that on different data sets, the optimal value range of C is from 2^{-4} to 2^{-1} , and on most datasets, OIFL obtains the best performance when the value of C is small enough. On the other hand, for scale factor σ , it has a better effect on the prediction performance of the model when its value is about 0.5. Besides, When C and σ are increased simultaneously, the performance of the classifier decreases significantly, the main reason may be caused by the overfitting of the model.

5.6. Different feature evolution analysis

To make our algorithm more persuasive, we study different modes of feature evolution on four data sets (air, dna, heart, and breast) and two real-world data streams (weather and covtypeNorm). We set B = 0.5 which means half of the features are used for learning at each round t and adjust δ to make the query ratio near 50%. For each data set,

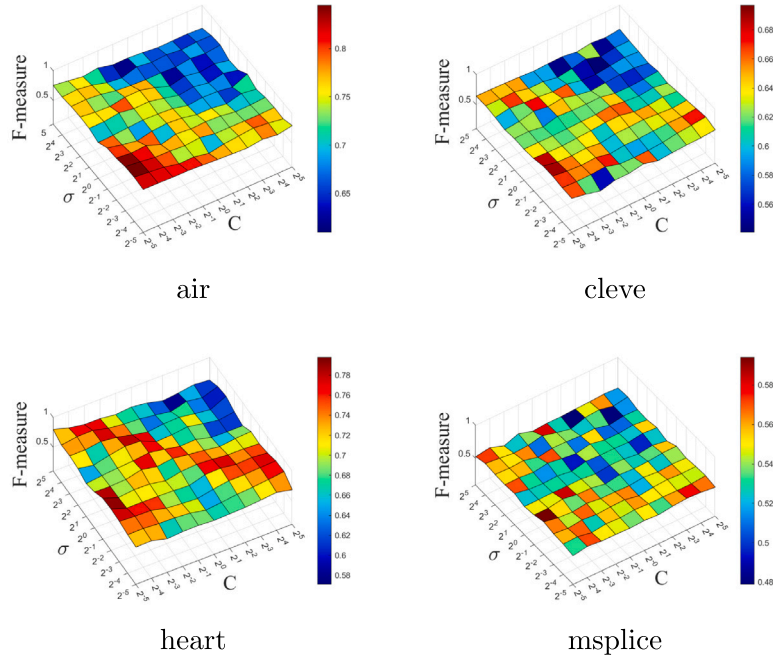


Fig. 4. Experimental F-measure results on four datasets(air, cleve, heart, and msplice) respect to parameter C and σ .

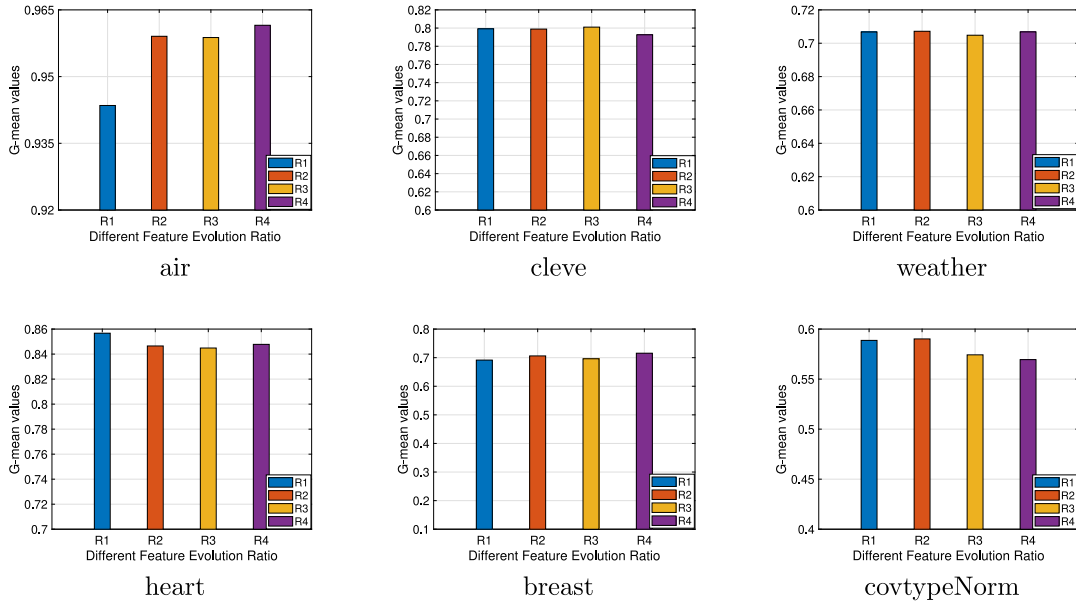


Fig. 5. Experimental G-mean results on four datasets(air, cleve, heart, and breast) and two real-world data streams (weather and covtypeNorm) respect to different modes of feature evolution.

we repeat the experiment 10 times and report average results, each with the same settings and divisions of data as general settings in S5.1. The results are reported by an average performance. Moreover, for ease of presentation, we employ notation to represent different feature evolution. (the notion R1 means that there are 30% features in S_1 and 70% features in S_2 . R2 indicates that S_1 and S_2 contain 40% features and 60% features, respectively. R3 has the same settings as above. R4 means that S_1 contains 70% features and the rest are in S_2 .)

As shown in Fig. 5, on the one hand, OIFL shows good performance for different feature evolution on the test data set, which verifies the effectiveness of the algorithm. On the other hand, we find that under the setting of different feature evolution in the same data set, the performance of our algorithm is not greatly affected, which reflects

that the algorithm shows good robustness to different feature evolution. The above results further reveal that our proposed method can accurately extract the important information in the data streams, and it can achieve satisfying results even when facing a more complex environment.

5.7. Dynamic imbalance ratio analysis

In this subsection, we apply OIFL on two synthetic datasets (dna and msplice) and two real-world data streams (weather and covtypeNorm) to assess the true value of the proposed algorithm. The general settings of the experiment are similar to that above. Table 5 shows the average AUC result of these seven methods with the query ratio near 10%

Table 5

Experimental AUC result (mean $\pm$ std(rank)) on the four datasets with the query ratio fixed to about 10% and 20% with different imbalance query ratios.

Data	Methods	AUC measures about requesting 10% labels			AUC measures about requesting 20% labels		
		$IR = 1$	$IR = 50$	$IR = 100$	$IR = 1$	$IR = 50$	$IR = 100$
dna	PEA	.759 $\pm$.040	.747 $\pm$.030	.718 $\pm$.034	.756 $\pm$.042	.740 $\pm$.037	.688 $\pm$.037
	RPEA	.737 $\pm$.044	.729 $\pm$.046	.716 $\pm$.047	.755 $\pm$.054	.734 $\pm$.060	.688 $\pm$.037
	FLLS	.746 $\pm$.034	.755 $\pm$.032	.752 $\pm$.039	.745 $\pm$.041	.753 $\pm$.042	.758 $\pm$.040
	RFLLS	.705 $\pm$.064	.727 $\pm$.065	.701 $\pm$.064	.722 $\pm$.038	.720 $\pm$.062	.703 $\pm$.052
	CPA	.796 $\pm$.025	.726 $\pm$.028	.719 $\pm$.025	.792 $\pm$.031	.713 $\pm$.022	.712 $\pm$.025
	PAUM	.738 $\pm$.039	.663 $\pm$.035	.646 $\pm$.028	.735 $\pm$.042	.676 $\pm$.044	.653 $\pm$.044
	OIFL	.914 $\pm$.018	.859 $\pm$.016	.858 $\pm$.019	.935 $\pm$.012	.904 $\pm$.018	.881 $\pm$.019
msplice	PEA	.777 $\pm$.043	.675 $\pm$.038	.738 $\pm$.041	.776 $\pm$.043	.652 $\pm$.045	.698 $\pm$.042
	RPEA	.744 $\pm$.043	.652 $\pm$.056	.710 $\pm$.049	.731 $\pm$ 0.060	.576 $\pm$.055	.666 $\pm$.068
	FLLS	.779 $\pm$.050	.671 $\pm$.024	.743 $\pm$.035	.761 $\pm$.052	.708 $\pm$.041	.746 $\pm$.043
	RFLLS	.711 $\pm$.059	.636 $\pm$.047	.711 $\pm$.045	.739 $\pm$.061	.632 $\pm$.066	.722 $\pm$.054
	CPA	.900 $\pm$.020	.786 $\pm$.034	.825 $\pm$.032	.893 $\pm$.025	.789 $\pm$.041	.799 $\pm$.032
	PAUM	.836 $\pm$.042	.699 $\pm$.056	.733 $\pm$.040	.837 $\pm$.047	.698 $\pm$.050	.748 $\pm$.045
	OIFL	.930 $\pm$.014	.890 $\pm$.028	.922 $\pm$.020	.966 $\pm$.012	.961 $\pm$.031	.975 $\pm$.011
covtypeNorm	PEA	.552 $\pm$.025	.553 $\pm$.017	.547 $\pm$.017	.541 $\pm$.020	.541 $\pm$.011	.560 $\pm$.029
	RPEA	.531 $\pm$.021	.543 $\pm$.011	.544 $\pm$.011	.536 $\pm$.016	.552 $\pm$.013	.555 $\pm$.023
	FLLS	.531 $\pm$.017	.549 $\pm$.015	.547 $\pm$.021	.535 $\pm$.023	.548 $\pm$.012	.554 $\pm$.016
	RFLLS	.530 $\pm$.032	.529 $\pm$.023	.538 $\pm$.018	.529 $\pm$.028	.527 $\pm$.027	.545 $\pm$.024
	CPA	.557 $\pm$.009	.562 $\pm$.012	.559 $\pm$.011	.554 $\pm$.017	.560 $\pm$.011	.562 $\pm$.011
	PAUM	.540 $\pm$.019	.555 $\pm$.020	.553 $\pm$.011	.530 $\pm$.010	.548 $\pm$.013	.556 $\pm$.011
	OIFL	.652 $\pm$.015	.601 $\pm$.014	.607 $\pm$.014	.651 $\pm$.011	.594 $\pm$.017	.591 $\pm$.019
weather	PEA	.677 $\pm$.030	.682 $\pm$.028	.671 $\pm$.035	.685 $\pm$.035	.680 $\pm$.015	.658 $\pm$.038
	RPEA	.682 $\pm$.022	.658 $\pm$.066	.676 $\pm$.032	.678 $\pm$ 0.40	.665 $\pm$.022	.668 $\pm$.034
	FLLS	.683 $\pm$.033	.675 $\pm$.023	.675 $\pm$.027	.682 $\pm$.021	.662 $\pm$.027	.667 $\pm$.030
	RFLLS	.675 $\pm$.033	.655 $\pm$.058	.671 $\pm$.032	.669 $\pm$.035	.673 $\pm$.033	.662 $\pm$.072
	CPA	.696 $\pm$.003	.688 $\pm$.003	.687 $\pm$.005	.698 $\pm$.008	.689 $\pm$.006	.686 $\pm$.006
	PAUM	.690 $\pm$.007	.681 $\pm$.006	.677 $\pm$.009	.689 $\pm$.006	.673 $\pm$.019	.666 $\pm$.015
	OIFL	.719 $\pm$.004	.699 $\pm$.006	.702 $\pm$.008	.720 $\pm$.004	.700 $\pm$.007	.700 $\pm$.007

and 20% in different imbalance ratios ($IR = 1$, $IR = 50$, $IR = 100$). The notion IR represents the imbalance ratio between negative and positive examples. The best AUC results are highlighted in boldface based on the paired t-tests at 95% significance level. We can find out that OIFL obtains the best average AUC values, which indicates OIFL would obtain good performance in real-world applications. Also, by comparing the performance between $IR = 50$ and $IR = 100$, we would obtain that our method is feebly affected as IR increases. This phenomenon means that OIFL can achieve robust results in highly and gradually changed imbalanced data distributions.

6. Conclusion

This paper explores a highly dynamic model from imbalanced data streams with label scarcity and proposes a new algorithm called Online Imbalance learning with unpredictable Feature evolution and Label scarcity (OIFL). The effectiveness of our algorithm in this complex scenario is tested empirically. We utilize margin-based online active learning to selectively label valuable instances. After obtaining the labels, we handle imbalanced class distribution by optimizing the F-measure and transforming its optimization into weighted surrogate loss minimization. When data streams arrive with augmented features, we combine the online passive-aggressive algorithm and structural risk minimization to update the classifier in the divided feature space. When data streams arrive with incomplete features, we leverage variance to identify the most informative features following the empirical risk minimization principle and continue to update the existing classifier as before. Experiments studies and theoretical analysis validate the effectiveness of our method.

The algorithm proposed in this paper is mainly for classification problems in linear scenes. Therefore, it is a very important work for us to extend the algorithm in this paper to nonlinear scenes in the future. Although our OIFL can work well with imbalanced data streams,

how to adapt to concept drift and mine higher-order information in imbalanced data streams is still a challenging and open problem in the above environments. Therefore, we will further study in the future work.

CRedit authorship contribution statement

Jiahang Tu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology. **Shilin Gu:** Writing – review & editing, Supervision, Methodology. **Chenping Hou:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Proof of Lemma 4.1

Proof. To prove the above inequality, we enumerate all the possible cases of Λ_t, H_t, R_t for discussion as follows:

Case 1: if $\Lambda_t = 0$, the inequality Eq. (21) holds with equality since $\mathbf{w}_t = \mathbf{w}_{t+1}$ and $\tau_t = 0$.

Case 2: if $\Lambda_t = 1$ and $R_t = 0$, we query the label and make right prediction $\hat{y}_t = y_t$. Also, there are two sub-cases:

sub-case 2.1: if $H_t = 0$ and $\ell_t(\mathbf{w}_t) = 0$, we have $\tau_t = 0, \mathbf{w}_t = \mathbf{w}_{t+1}$, which similar to Case 1.

sub-case 2.2: if $H_t = 1$ and $\ell_t(\mathbf{w}_t) > 0$, we have $\bar{\mathbf{w}}_{t+1} = [\mathbf{w}_t + \tau_t \gamma_t \mathbf{x}_t, \tau_t \gamma_t \Pi_{\bar{\mathbf{w}}_t} \mathbf{x}_t]$, and since $\mathbf{w}_{t+1}$ is the truncated version of $\bar{\mathbf{w}}_{t+1}$, we would get $\mathbf{w}_{t+1} \leq \bar{\mathbf{w}}_{t+1}$ and $\|\mathbf{w}_{t+1}\|^2 \leq \|\bar{\mathbf{w}}_{t+1}\|^2$. Furthermore, from

$$\begin{aligned} & \|\mathbf{w}_{t+1} - \kappa \Pi_{\mathbf{w}_{t+1}} \mathbf{c}\|^2 - \|\bar{\mathbf{w}}_{t+1} - \kappa \Pi_{\bar{\mathbf{w}}_{t+1}} \mathbf{c}\|^2 \\ &= \kappa^2 \|\mathbf{w}_{t+1}\|^2 - \kappa^2 \|\bar{\mathbf{w}}_{t+1}\|^2 - 2\kappa \|\Pi_{\mathbf{w}_{t+1}} \mathbf{c}\|(\|\bar{\mathbf{w}}_{t+1}\| - \|\mathbf{w}_{t+1}\|) \leq 0, \end{aligned} \quad (\text{A.1})$$

we have:

$$\|\mathbf{w}_{t+1} - \kappa \Pi_{\mathbf{w}_{t+1}} \mathbf{c}\|^2 \leq \|\bar{\mathbf{w}}_{t+1} - \kappa \Pi_{\bar{\mathbf{w}}_{t+1}} \mathbf{c}\|^2. \quad (\text{A.2})$$

Then, for $t \in [1, T_1]$

$$\begin{aligned} & \|\mathbf{w}_t - \kappa \Pi_{\mathbf{w}_t} \mathbf{c}\|^2 - \|\mathbf{w}_{t+1} - \kappa \Pi_{\mathbf{w}_{t+1}} \mathbf{c}\|^2 \\ & \geq \|\mathbf{w}_t - \kappa \Pi_{\mathbf{w}_t} \mathbf{c}\|^2 - \|\bar{\mathbf{w}}_{t+1} - \kappa \Pi_{\bar{\mathbf{w}}_{t+1}} \mathbf{c}\|^2 \\ &= \|\mathbf{w}_t - \kappa \Pi_{\mathbf{w}_t} \mathbf{c}\|^2 - \|\mathbf{w}_t + \tau_t \gamma_t \mathbf{x}_t - \kappa \Pi_{\mathbf{w}_t} \mathbf{c}\|^2 \\ & \quad - \|\tau_t \gamma_t \mathbf{x}_t - \kappa \Pi_{\mathbf{w}_{t+1}/\mathbf{w}_t} \mathbf{c}\|^2 \\ &= 2\tau_t \gamma_t \mathbf{x}_t (\kappa \Pi_{\mathbf{w}_t} \mathbf{c} - \mathbf{w}_t) \cdot \Pi_{\mathbf{w}_t} \mathbf{x}_t - (\tau_t \gamma_t)^2 \|\Pi_{\mathbf{w}_t} \mathbf{x}_t\|^2 \\ & \quad - \|\tau_t \gamma_t \mathbf{x}_t - \kappa \Pi_{\mathbf{w}_{t+1}/\mathbf{w}_t} \mathbf{c}\|^2 \\ &= 2\gamma_t \tau_t (y_t \kappa \Pi_{\mathbf{w}_t} \mathbf{c} \cdot \Pi_{\mathbf{w}_t} \mathbf{x}_t + y_t \kappa \Pi_{\mathbf{w}_{t+1}/\mathbf{w}_t} \mathbf{c} \cdot \Pi_{\bar{\mathbf{w}}_t} \mathbf{x}_t) \\ & \quad - (\tau_t \gamma_t)^2 \|\mathbf{x}_t\|^2 - 2\tau_t \gamma_t \mathbf{x}_t \cdot \Pi_{\mathbf{w}_t} \mathbf{x}_t - \kappa^2 \|\Pi_{\mathbf{w}_{t+1}/\mathbf{w}_t} \mathbf{c}\|^2. \end{aligned} \quad (\text{A.3})$$

By definition, $\|\Pi_{\mathbf{w}_t} \mathbf{x}_t\|^2 + \|\Pi_{\mathbf{w}_{t+1}/\mathbf{w}_t} \mathbf{x}_t\|^2 = \|\mathbf{x}_t\|^2$. Since $\ell_t^*(\mathbf{c}) \geq 1 - y_t (\Pi_{\mathbf{x}_t} \mathbf{c} \cdot \mathbf{x}_t)$, we get $\ell_t^*(\mathbf{c}) \geq 1 - y_t \Pi_{\mathbf{w}_t} \mathbf{c} \cdot \mathbf{x}_t - y_t \Pi_{\bar{\mathbf{w}}_t} \mathbf{c} \cdot \mathbf{x}_t$. Then, we have

$$\begin{aligned} P_1 &= \|\mathbf{w}_t - \kappa \Pi_{\mathbf{w}_t} \mathbf{c}\|^2 - \|\mathbf{w}_{t+1} - \kappa \Pi_{\mathbf{w}_{t+1}} \mathbf{c}\|^2 + \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 \\ & \quad + 2\gamma_t \tau_t \kappa \ell_t^*(\mathbf{c}) + \kappa^2 \|\Pi_{\mathbf{w}_{t+1}/\mathbf{w}_t} \mathbf{c}\|^2 \\ & \geq 2\gamma_t \tau_t (\kappa - y_t \mathbf{w}_t \cdot \Pi_{\mathbf{w}_t} \mathbf{x}_t). \end{aligned} \quad (\text{A.4})$$

Similar to the above deduction, for $t \in (T_1, T_2]$, we have the inequality

$$\begin{aligned} & \|\mathbf{w}_t - \kappa \Pi_{\mathbf{w}_t} \mathbf{c}\|^2 - \|\mathbf{w}_{t+1} - \kappa \Pi_{\mathbf{w}_{t+1}} \mathbf{c}\|^2 + \tau_t^2 \gamma_t^2 [(m_t^s)^2 \|\mathbf{x}_t^s\|^2 + (m_t^n)^2 \|\mathbf{x}_t^n\|^2] \\ & \quad + 2\gamma_t \tau_t \kappa \ell_t^*(\mathbf{c}) \\ & \geq 2\gamma_t \tau_t (\kappa - m_t^s y_t \mathbf{w}_t^s \cdot \mathbf{x}_t^s - m_t^n y_t \mathbf{w}_t^n \cdot \mathbf{x}_t^n). \end{aligned} \quad (\text{A.5})$$

Since $m_t^s, m_t^n \in [0, 1]$ and $m_t^s + m_t^n \leq 1$, we have $\|\mathbf{x}_t\|^2 \geq (m_t^s)^2 \|\mathbf{x}_t^s\|^2 + (m_t^n)^2 \|\mathbf{x}_t^n\|^2$. Therefore, inequality Eq. (A.5) holds:

$$\begin{aligned} P_2 &= \|\mathbf{w}_t - \kappa \Pi_{\mathbf{w}_t} \mathbf{c}\|^2 - \|\mathbf{w}_{t+1} - \kappa \Pi_{\mathbf{w}_{t+1}} \mathbf{c}\|^2 + \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 + 2\gamma_t \tau_t \kappa \ell_t^*(\mathbf{c}) \\ & \geq 2\gamma_t \tau_t (\kappa - m_t^s y_t \mathbf{w}_t^s \cdot \mathbf{x}_t^s - m_t^n y_t \mathbf{w}_t^n \cdot \mathbf{x}_t^n). \end{aligned} \quad (\text{A.6})$$

On account of $R_t = 0$ and $H_t = 1$, which means that $0 < 1 - y_t \Pi_{\mathbf{x}_t} \mathbf{w}_t \cdot \mathbf{x}_t < 1$. Hence, inequality Eqs. (A.4) and (A.6) holds:

$$P_1 \geq 2\gamma_t \tau_t (\kappa - |p_t|), \quad P_2 \geq 2\gamma_t \tau_t (\kappa - |p_t|). \quad (\text{A.7})$$

Case 3: if $\Lambda_t = 1$ and $R_t = 1$, we obtain the label of $\mathbf{x}_t$ while the predicted label $\hat{y}_t \neq y_t$, also, $H_t = 0$. We have the same P_1 and P_2 as above. Since $R_t = 1$ implies $y_t \Pi_{\mathbf{x}_t} \mathbf{w}_t \cdot \mathbf{x}_t \leq 0$ and $-y_t \Pi_{\mathbf{x}_t} \mathbf{w}_t \cdot \mathbf{x}_t = |q_t|$. Therefore, inequality Eqs. (A.4) and (A.6) holds:

$$P_1 \geq 2\gamma_t \tau_t (\kappa + |p_t|), \quad P_2 \geq 2\gamma_t \tau_t (\kappa + |p_t|). \quad (\text{A.8})$$

Summarizing all the above cases of different phases for all $t = 1, 2, \dots, T$.

Firstly, for $t \in [1, T_1]$, we have,

$$\begin{aligned} & \sum_{t=1}^{T_1} (2H_t \Lambda_t \gamma_t \tau_t (\kappa - |p_t|) + 2R_t \Lambda_t \gamma_t \tau_t (\kappa + |p_t|)) \\ & \leq \sum_{t=1}^{T_1} P_1 = \sum_{t=1}^{T_1} (\|\mathbf{w}_t - \kappa \Pi_{\mathbf{w}_t} \mathbf{c}\|^2 - \|\mathbf{w}_{t+1} - \kappa \Pi_{\mathbf{w}_{t+1}} \mathbf{c}\|^2) \\ & \quad + \sum_{t=1}^{T_1} (\tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 + 2\gamma_t \tau_t \kappa \ell_t^*(\mathbf{c})) + \sum_{t=1}^{T_1} (\kappa^2 \|\Pi_{\mathbf{w}_{t+1}/\mathbf{w}_t} \mathbf{c}\|^2). \end{aligned} \quad (\text{A.9})$$

Then, similar to P_1 , for $t \in (T_1, T_2]$, we have

$$\begin{aligned} & \sum_{t=T_1+1}^{T_2} (2H_t \Lambda_t \gamma_t \tau_t (\kappa - |p_t|) + 2R_t \Lambda_t \gamma_t \tau_t (\kappa + |p_t|)) \\ & \leq \sum_{t=T_1+1}^{T_2} P_2 = \sum_{t=T_1+1}^{T_2} (\|\mathbf{w}_t - \kappa \Pi_{\mathbf{w}_t} \mathbf{c}\|^2 - \|\mathbf{w}_{t+1} - \kappa \Pi_{\mathbf{w}_{t+1}} \mathbf{c}\|^2) \\ & \quad + \sum_{t=T_1+1}^{T_2} \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 + \sum_{t=T_1+1}^{T_2} 2\gamma_t \tau_t \kappa \ell_t^*(\mathbf{c}). \end{aligned} \quad (\text{A.10})$$

Combining the inequality Eqs. (A.9) and (A.10), we would get the inequality:

$$\begin{aligned} & \sum_{t=1}^T (2H_t \Lambda_t \gamma_t \tau_t (\kappa - |p_t|) + 2R_t \Lambda_t \gamma_t \tau_t (\kappa + |p_t|)) \\ & \leq \sum_{t=1}^T (\|\mathbf{w}_t - \kappa \Pi_{\mathbf{w}_t} \mathbf{c}\|^2 - \|\mathbf{w}_{t+1} - \kappa \Pi_{\mathbf{w}_{t+1}} \mathbf{c}\|^2) \\ & \quad + \sum_{t=1}^T \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 + \sum_{t=1}^T 2\gamma_t \tau_t \kappa \ell_t^*(\mathbf{c}) + \sum_{t=1}^{T_1} (\kappa^2 \|\Pi_{\mathbf{w}_{t+1}/\mathbf{w}_t} \mathbf{c}\|^2) \\ &= \|\mathbf{w}_1 - \kappa \Pi_{\mathbf{w}_1} \mathbf{c}\|^2 - \|\mathbf{w}_{T+1} - \kappa \Pi_{\mathbf{w}_{T+1}} \mathbf{c}\|^2 \\ & \quad + \sum_{t=1}^T \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 + \sum_{t=1}^T 2\gamma_t \tau_t \kappa \ell_t^*(\mathbf{c}) + \sum_{t=1}^{T_1} (\kappa^2 \|\Pi_{\mathbf{w}_{t+1}/\mathbf{w}_t} \mathbf{c}\|^2). \end{aligned} \quad (\text{A.11})$$

Since $\mathbf{w}_1 = (0, \dots, 0) \in \mathbb{R}^{d_1}$, $\|\mathbf{w}_{T+1} - \kappa \Pi_{\mathbf{w}_{T+1}} \mathbf{c}\|^2 \geq 0$. Also, $\Pi_{\mathbf{w}_1} \mathbf{c} = \Pi_{\mathbf{w}_1^s} \mathbf{c}$, which shares the same feature space. we have:

$$\begin{aligned} & \sum_{t=1}^T 2\Lambda_t \tau_t \gamma_t [H_t (\kappa - |p_t|) + R_t (\kappa + |p_t|)] \\ & \leq \kappa^2 \|\Pi_{\mathbf{w}_1} \mathbf{c}\|^2 + \sum_{t=1}^{T_1} (\kappa^2 \|\Pi_{\mathbf{w}_{t+1}/\mathbf{w}_t} \mathbf{c}\|^2) + \sum_{t=1}^T \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 \\ & \quad + \sum_{t=1}^T 2\kappa \tau_t \gamma_t \ell_t^*(\mathbf{c}). \\ & \leq \kappa^2 \|\mathbf{c}\|^2 + \sum_{t=1}^T \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 + \sum_{t=1}^T 2\kappa \tau_t \gamma_t \ell_t^*(\mathbf{c}). \end{aligned} \quad (\text{A.12})$$

The lemma is proved. $\square$

Appendix B. Proof of Theorem 1

Proof. Bringing assume $\ell_t^*(\mathbf{c}) = 0$ into the inference of Lemma 4.1, we can get

$$\sum_{t=1}^T 2\Lambda_t \tau_t \gamma_t [H_t (\kappa - |p_t|) + R_t (\kappa + |p_t|)] \leq \kappa^2 \|\mathbf{c}\|^2 + \sum_{t=1}^T \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2. \quad (\text{B.1})$$

Furthermore, the inequality can be deduced as:

$$\begin{aligned}
\kappa^2 \|c\|^2 &\geq \sum_{t=1}^T 2A_t \tau_t \gamma_t [H_t(\kappa - |p_t|) + R_t(\kappa + |p_t|)] - \sum_{t=1}^T \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 \\
&= \sum_{t=1}^T 2A_t \tau_t \gamma_t [H_t(\kappa - |p_t| - \frac{\tau_t \gamma_t}{2} \|\mathbf{x}_t\|^2)] \\
&\quad + \sum_{t=1}^T 2A_t \tau_t \gamma_t [R_t(\kappa + |p_t| - \frac{\tau_t \gamma_t}{2} \|\mathbf{x}_t\|^2)] \\
&\geq \sum_{t=1}^T 2A_t \tau_t \gamma_t [H_t(\kappa - |p_t| - \frac{\ell_t(\mathbf{w}_t)}{2})] \\
&\quad + \sum_{t=1}^T 2A_t \tau_t \gamma_t [R_t(\kappa + |p_t| - \frac{\ell_t(\mathbf{w}_t)}{2})] \\
&= \sum_{t=1}^{T_1} 2A_t \tau_t \gamma_t [H_t(\kappa - \frac{1+|p_t|}{2})] \\
&\quad + \sum_{t=1}^{T_1} 2A_t \tau_t \gamma_t [R_t(\kappa - \frac{1-|p_t|}{2})] \\
&\quad + \sum_{t=T_1+1}^{T_2} 2A_t \tau_t \gamma_t [H_t(\kappa - |p_t| \\
&\quad - \frac{1 - y_t(m_t^s \Pi_{\mathbf{x}_t^s} \mathbf{w}_t \cdot \mathbf{x}_t^s + m_t^n \Pi_{\mathbf{x}_t^n} \mathbf{w}_t \cdot \mathbf{x}_t^n)}{2})] \\
&\quad + \sum_{t=T_1+1}^{T_2} 2A_t \tau_t \gamma_t [R_t(\kappa + |p_t| \\
&\quad - \frac{1 - y_t(m_t^s \Pi_{\mathbf{x}_t^s} \mathbf{w}_t \cdot \mathbf{x}_t^s + m_t^n \Pi_{\mathbf{x}_t^n} \mathbf{w}_t \cdot \mathbf{x}_t^n)}{2})].
\end{aligned} \tag{B.2}$$

Put $m_t = \min\{m_t^s, m_t^n\}$ and $m_t \in (0, 1)$ into Eq. (B.2), we have

$$\begin{aligned}
\kappa^2 \|c\|^2 &\geq \sum_{t=1}^{T_1} 2A_t \tau_t \gamma_t [H_t(\kappa - \frac{1+|p_t|}{2})] + \sum_{t=1}^{T_1} 2A_t \tau_t \gamma_t [R_t(\kappa - \frac{1-|p_t|}{2})] \\
&\quad + \sum_{t=T_1+1}^{T_2} 2A_t \tau_t \gamma_t [H_t(\kappa - |p_t| - \frac{1-m_t|p_t|}{2})] \\
&\quad + \sum_{t=T_1+1}^{T_2} 2A_t \tau_t \gamma_t [R_t(\kappa + |p_t| - \frac{(1-m_t|p_t|)}{2})] \\
&\geq \sum_{t=1}^{T_1} 2A_t \tau_t \gamma_t [H_t(\kappa - \frac{1+|p_t|}{2})] + \sum_{t=1}^{T_1} 2A_t \tau_t \gamma_t [R_t(\kappa - \frac{1-|p_t|}{2})] \\
&\quad + \sum_{t=T_1+1}^{T_2} 2A_t \tau_t \gamma_t [H_t(\kappa - \frac{1+(2-m_t)|p_t|}{2})] \\
&\quad + \sum_{t=T_1+1}^{T_2} 2A_t \tau_t \gamma_t [R_t(\kappa + \frac{((2+m_t)|p_t| - 1)}{2})] \\
&\geq \sum_{t=1}^T 2A_t \tau_t \gamma_t [H_t(\kappa - \frac{1}{2} - |p_t|)] + \sum_{t=1}^T 2A_t \tau_t \gamma_t [R_t(\kappa - \frac{1-|p_t|}{2})].
\end{aligned} \tag{B.3}$$

Plugging $\kappa = \frac{1+\nu}{2}$, $\nu \geq 1$ into the above inequality, we have

$$(\frac{1+\nu}{2})^2 \|c\|^2 \geq \sum_{t=1}^T A_t \tau_t \gamma_t [R_t(\nu + |p_t|)]. \tag{B.4}$$

When $R_t = 1$, it implies that the classifier $\mathbf{w}_t \in \mathbb{R}^{d_t}$ makes a wrong prediction, then we have $y_t(\Pi_{\mathbf{x}_t} \mathbf{w}_t \cdot \mathbf{x}_t) \leq 0$ and $\ell_t^*(\mathbf{w}_t) \geq 1$. Thus, we get $\tau_t \gamma_t \geq \min\{C\gamma_t, \frac{1}{G^2}\}$. Further, the above inequality can be reformulated

as:

$$\begin{aligned}
(\frac{1+\nu}{2})^2 \|c\|^2 &\geq \min\{C\gamma_t, \frac{1}{G^2}\} E[\sum_{t=1}^T R_t] E[A_t], \\
E[\sum_{t=1}^T R_t] &\leq \rho (\frac{1+\nu}{2})^2 \|c\|^2, \rho = \frac{1}{\nu} \min\{\frac{1}{C\gamma_t}, G^2\}.
\end{aligned} \tag{B.5}$$

The theorem is proved. $\square$

Appendix C. Proof of Theorem 2

Proof. As to Theorem 1, based on Lemma 4.1, we have

$$\begin{aligned}
\sum_{t=1}^T 2A_t \tau_t \gamma_t [H_t(\kappa - |p_t|) + R_t(\kappa + |p_t|)] &\leq \kappa^2 \|c\|^2 + \sum_{t=1}^T \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 \\
&\quad + \sum_{t=1}^T 2\kappa \tau_t \gamma_t \ell_t^*(c).
\end{aligned} \tag{C.1}$$

Furthermore, the inequality can be reformulated:

$$\begin{aligned}
\kappa^2 \|c\|^2 + \sum_{t=1}^T 2\kappa \tau_t \gamma_t \ell_t^*(c) &\geq - \sum_{t=1}^T \tau_t^2 \gamma_t^2 \|\mathbf{x}_t\|^2 \\
&\quad + \sum_{t=1}^T 2A_t \tau_t \gamma_t [H_t(\kappa - |p_t|) + R_t(\kappa + |p_t|)].
\end{aligned} \tag{C.2}$$

Similarly to the derivative process of Eqs. (B.2) and (B.3), the inequality holds:

$$\begin{aligned}
\kappa^2 \|c\|^2 + \sum_{t=1}^T 2\kappa \tau_t \gamma_t \ell_t^*(c) &\geq \sum_{t=1}^T 2A_t \tau_t \gamma_t [H_t(\kappa - \frac{1}{2} - |p_t|)] \\
&\quad + \sum_{t=1}^T 2A_t \tau_t \gamma_t [R_t(\kappa - \frac{1-|p_t|}{2})].
\end{aligned} \tag{C.3}$$

Plugging $\kappa = \frac{1+\nu}{2}$, $\nu \geq 1$ into the above inequality, we have

$$(\frac{1+\nu}{2})^2 \|c\|^2 + \sum_{t=1}^T (\nu + 1) \tau_t \gamma_t \ell_t^*(c) \geq \sum_{t=1}^T A_t \tau_t \gamma_t [R_t(\nu + |p_t|)]. \tag{C.4}$$

When $R_t = 1$, it implies that the classifier $\mathbf{w}_t \in \mathbb{R}^{d_t}$ makes a wrong prediction, then we have $y_t(\Pi_{\mathbf{x}_t} \mathbf{w}_t \cdot \mathbf{x}_t) \leq 0$ and $\ell_t^*(\mathbf{w}_t) \geq 1$. Thus, we get $\tau_t \gamma_t \geq \min\{C\gamma_t, \frac{1}{G^2}\}$. Further, combine the expectation with the above equation and rearrange the results to conclude, the above inequality can be reformulated as:

$$\begin{aligned}
E[\sum_{t=1}^T R_t] &\leq \rho (\frac{1+\nu}{2})^2 \|c\|^2 + \sum_{t=1}^T (\nu + 1) C\gamma_t \ell_t^*(c), \\
\rho &= \frac{1}{\nu} \min\{\frac{1}{C\gamma_t}, G^2\}.
\end{aligned} \tag{C.5}$$

The theorem is proved. $\square$

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